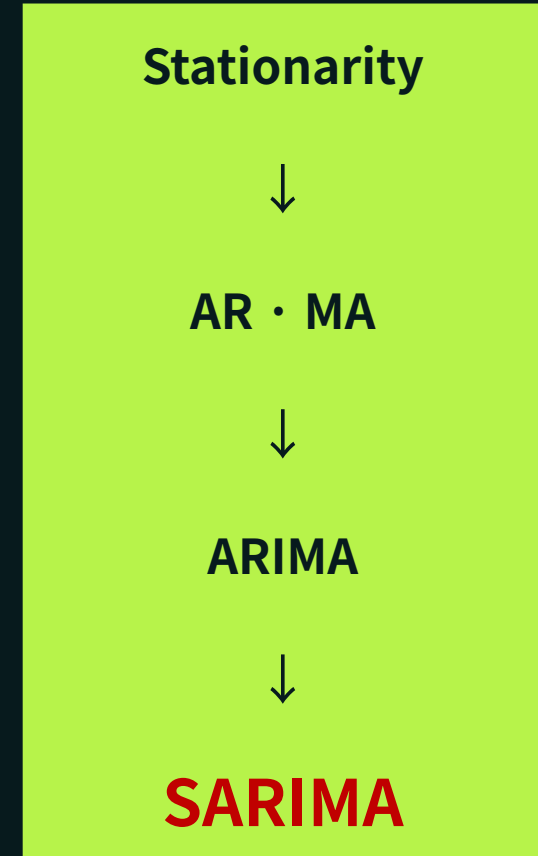


From Stationarity to Seasonal ARIMA

AR · MA · ARMA · ARIMA · SARIMA in one storyline



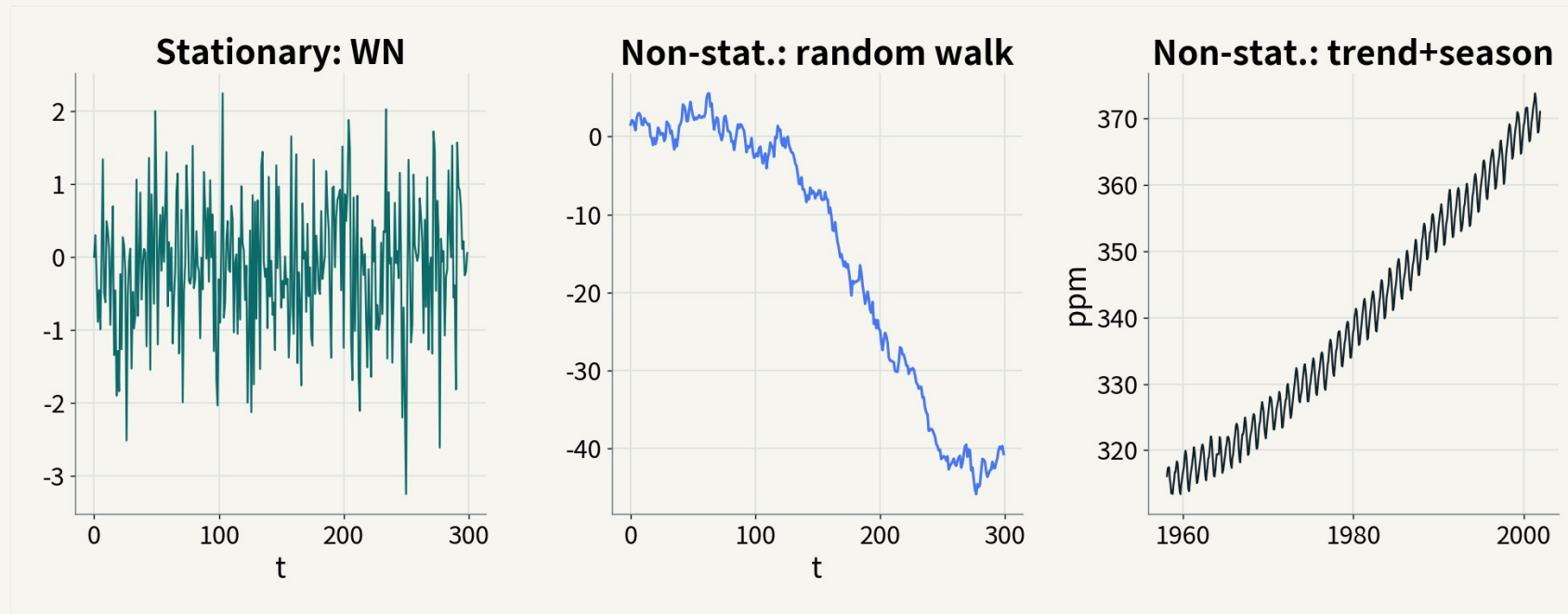
We build the model one layer at a time

01	Stationarity	The premise strict vs weak · differencing · volatility
02	Autoregressive AR	Uses past values AR(1) · long memory · $ \phi < 1$ · half-life
03	Moving Average MA	Uses past errors MA(1) · short memory · bid-ask bounce
04	ARMA · ARIMA	Combine, then difference ACF/PACF · auto selection · Box-Jenkins
05	Seasonal ARIMA	Look one season back (P,D,Q) _s · multiplicative · airline

stationarity → past values → past errors → combine & difference → season

Time-series observations are not independent — they need their own rules

Definition · observations recorded in time order $\{Y_1, Y_2, \dots, Y_t\}$



Ordinary regression assumes independence; in time series, yesterday shapes today
So first we must pin down what does NOT change over time

Five symbols carry the whole deck

Notation

Y_t : value at t , Y_{t-k} : lag k , $e_t \sim \text{WN}(0, \sigma^2)$

ω

Intercept

sets the mean

ϕ

AR coefficient

weight on values

θ

MA coefficient

weight on errors

e_t

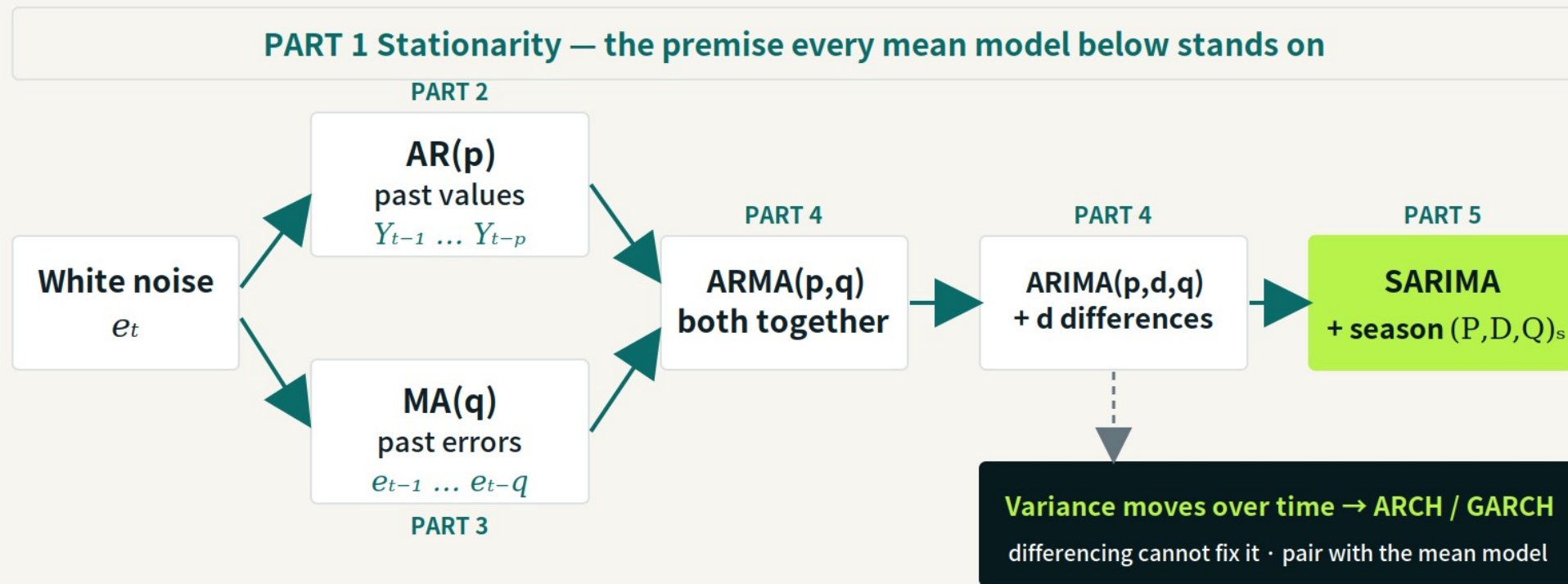
White noise

mean 0, no corr.

White noise e_t is the unpredictable news (shock)

The goal: leave nothing but white noise in the residuals

All five parts sit on one model map



※ The main line models the conditional mean; ARCH/GARCH models the variance on top of it

01

Stationarity — what does not move

To learn the future from the past, some rules must stay fixed

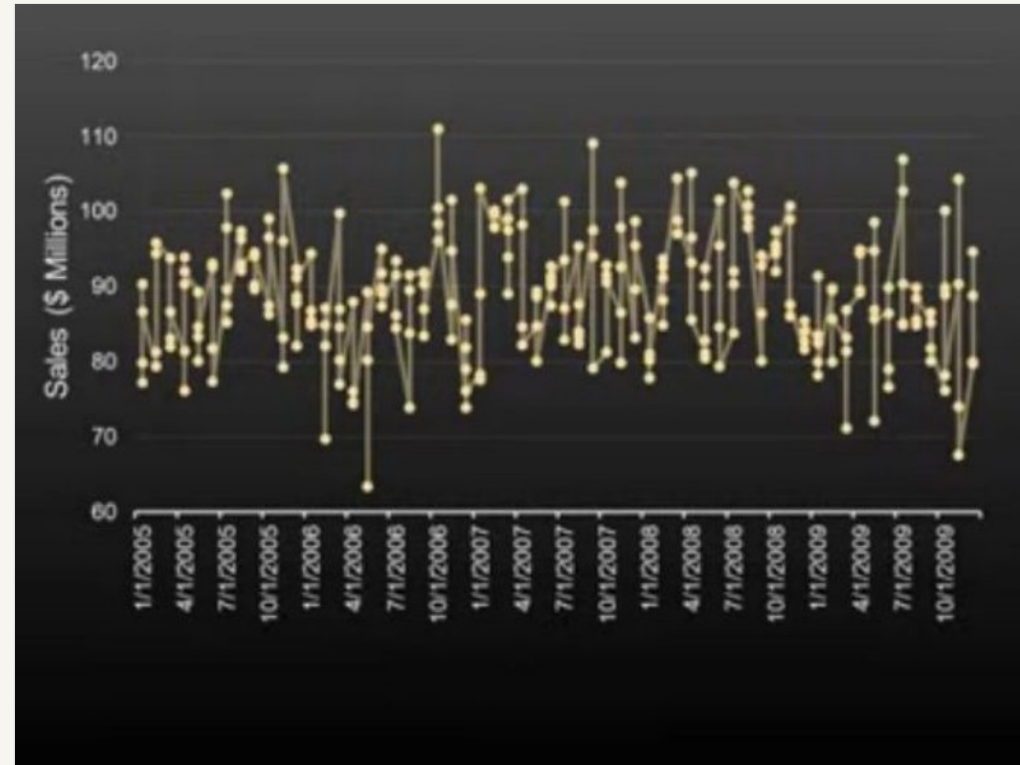
Stationarity means any window of the series has the same distribution

DEFINITION

Law depends only on the lag

- independence dropped
- consistency required
- calendar date irrelevant

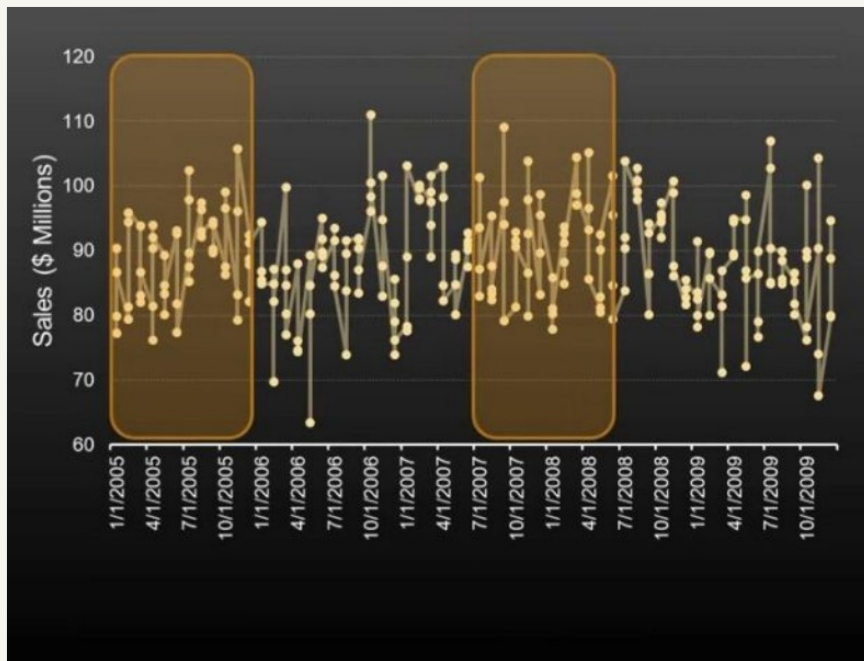
That is why past rules carry into the future



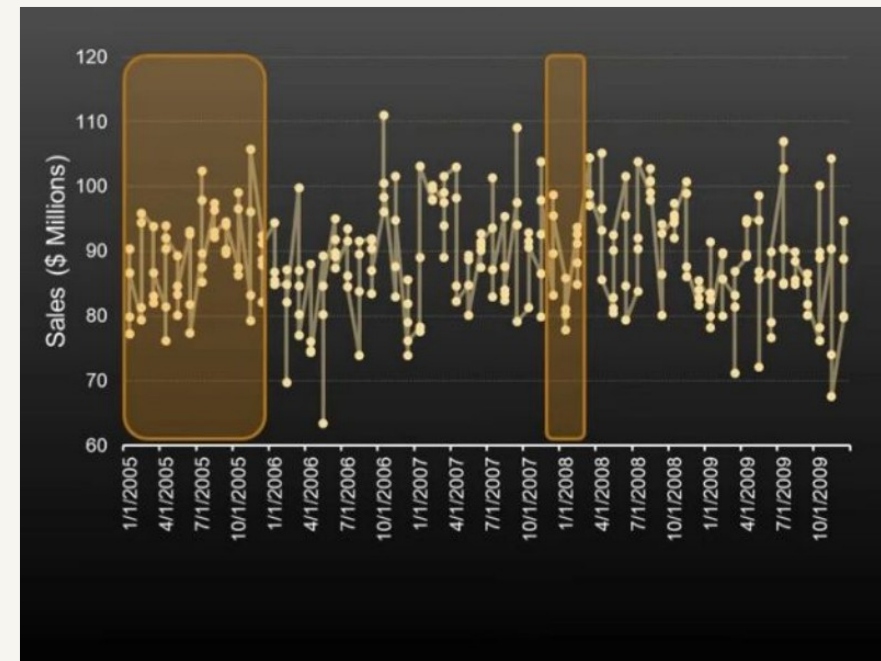
Stationarity = the probability law does not change as time passes

Compare windows of the same width — their location may differ

① Same width → same law



② Different widths → different laws



The law depends on how far apart (lag h), not on when (t)

Strict fixes the whole law; weak fixes only the first two moments

Strict — joint law invariant to time shifts

$$(Y_{t_1}, \dots, Y_{t_k}) \stackrel{d}{=} (Y_{t_1+h}, \dots, Y_{t_k+h}) \quad \forall k, h$$

Weak (covariance) — mean, variance, autocovariance invariant

$$E[Y_t] = \mu, \quad \text{Var}(Y_t) = \sigma^2 < \infty, \quad \text{Cov}(Y_t, Y_{t-h}) = \gamma(h)$$

As the lecture puts it: 'Weak will do!' — ARIMA needs only weak stationarity

In practice, plain 'stationary' usually means weakly stationary

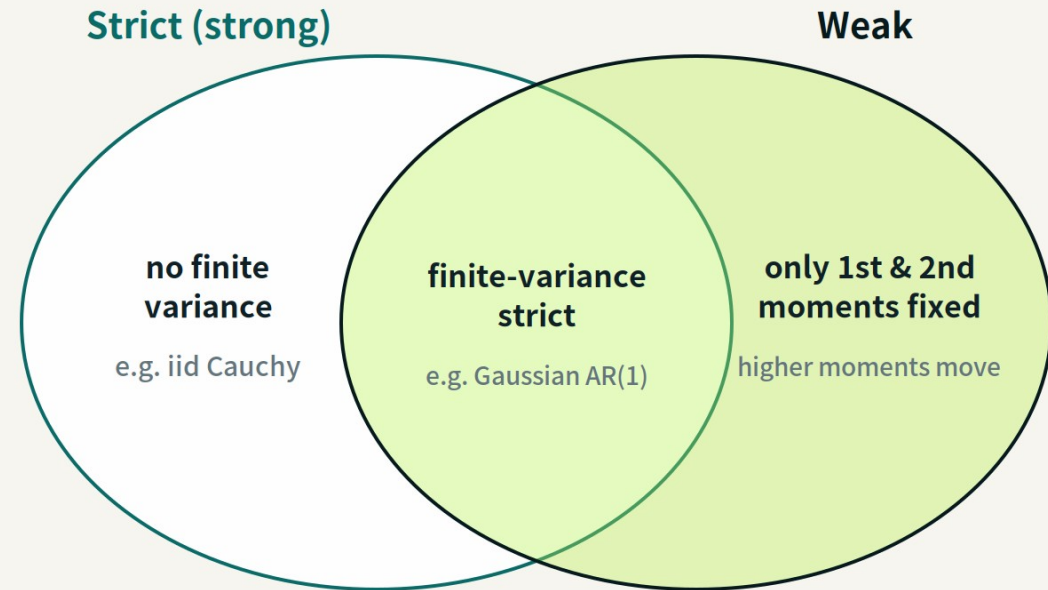
Strict stationarity does NOT imply weak stationarity

KEY POINT

Finite variance decides

- weak: finite variance
- strict: no such condition
- Gaussian: identical

'strong does NOT imply weak'



For Gaussian processes the two coincide

Whether the variance exists is what separates the two

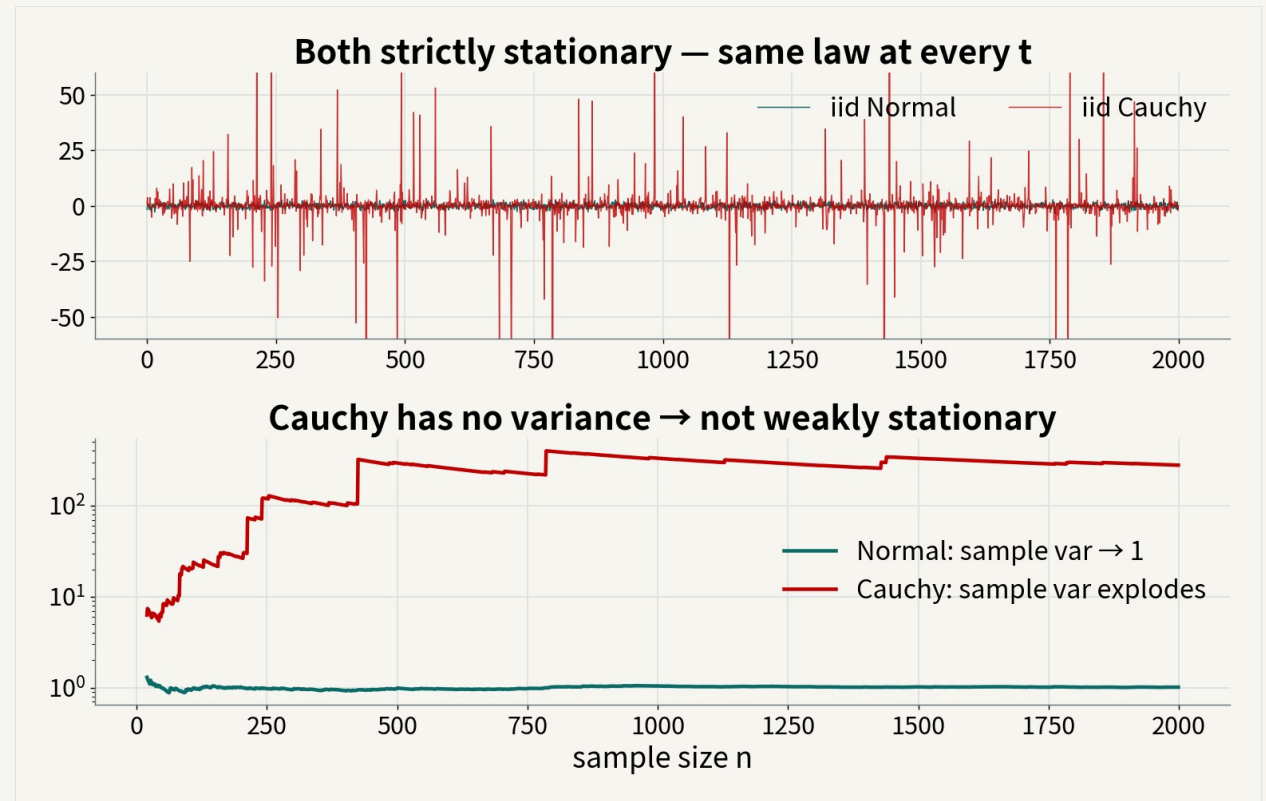
An iid Cauchy series is the classic counter-example

COUNTER-EXAMPLE

Cauchy law

- same law at every t
- no mean, no variance
- sample var. diverges

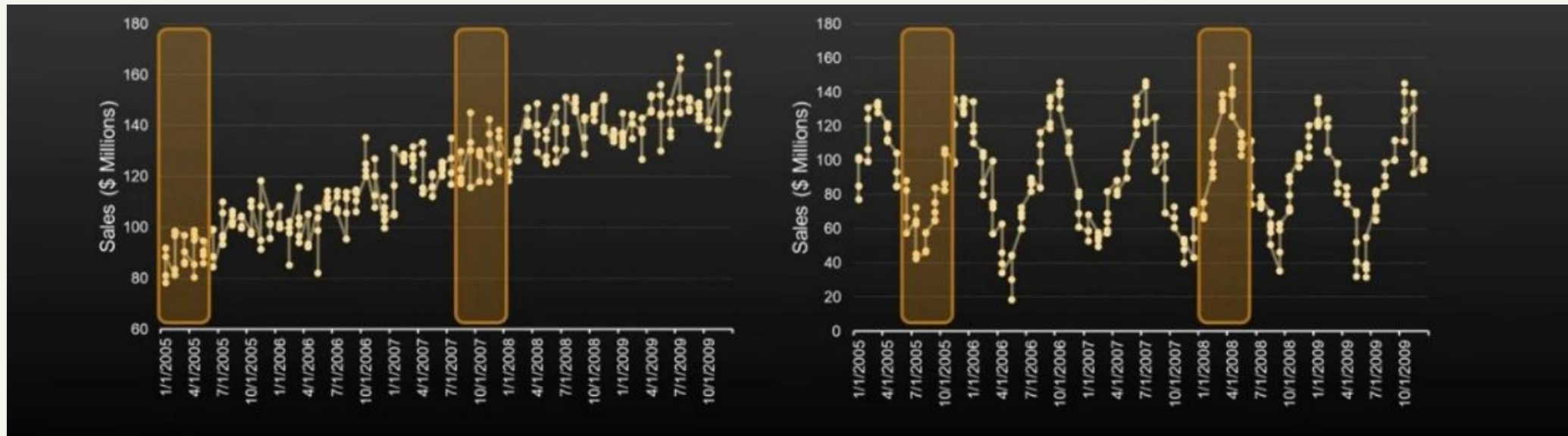
Fat tails: check variance first



2,000 draws · Normal settles at 1, Cauchy explodes

Trend and seasonality move the mean and break stationarity

Lecture figure · same-width windows, different levels



TREND

Mean keeps rising

SEASONALITY

Mean cycles

VERDICT

NOT stationary

Shift the window and the mean changes — past averages say nothing about the future

Test, don't eyeball — ADF and KPSS have opposite null hypotheses

Random walk (unit root): variance grows with t

$$Y_t = Y_{t-1} + e_t \Rightarrow \text{Var}(Y_t) = t \sigma^2$$

TEST	NULL H_0	WHITE NOISE p	RANDOM WALK p	CO ₂ p
ADF	unit root (non-stat.)	< 0.001	≥ 0.10	≥ 0.10
KPSS	stationary	≥ 0.10	≤ 0.01	≤ 0.01

※ KPSS p-values are truncated at table limits (0.01–0.10) · simulated series and monthly CO₂

Stationary = ADF rejects AND KPSS does not reject

If they disagree, decide on differencing with extra care

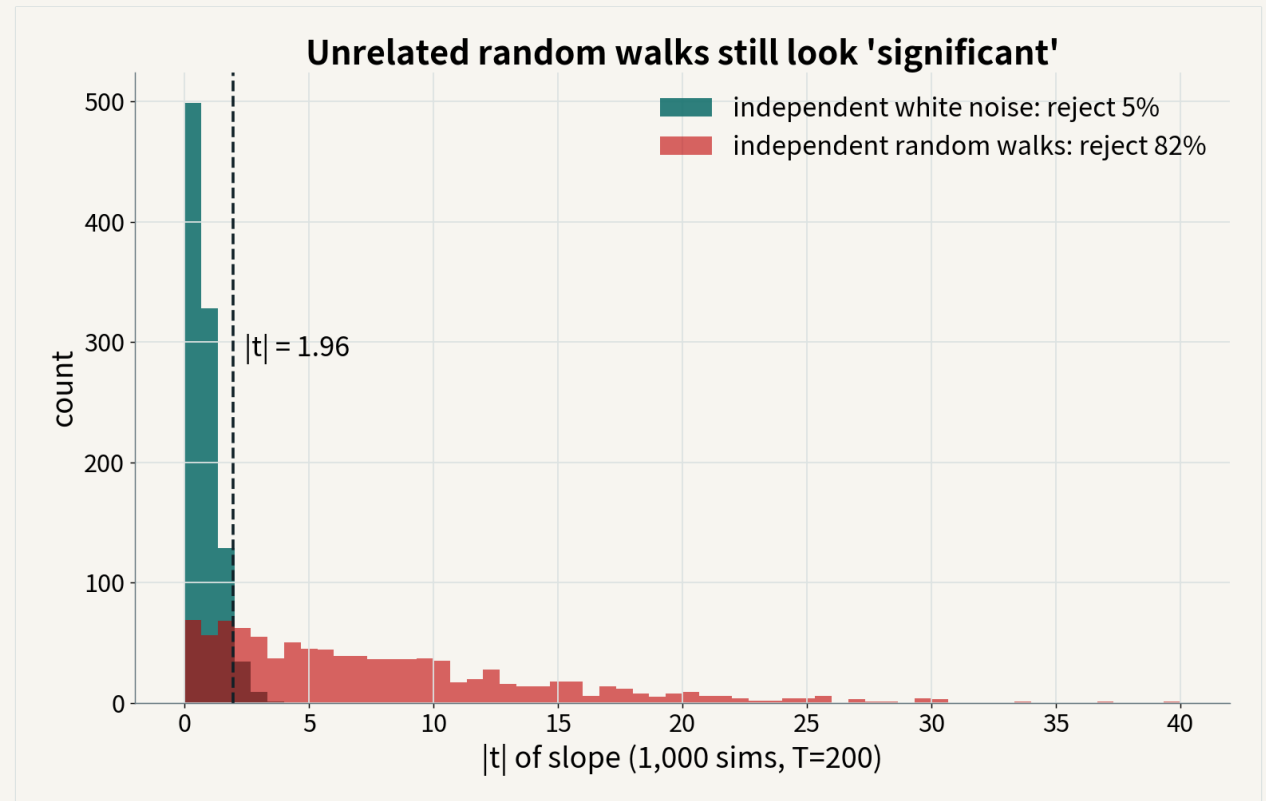
Regressing non-stationary series manufactures 'significant' links

**EPISODE · Granger–Newbold
1974**

Spurious regression

- two unrelated random walks
- 82% flagged at 5%
- median R^2 0.16

Check stationarity first



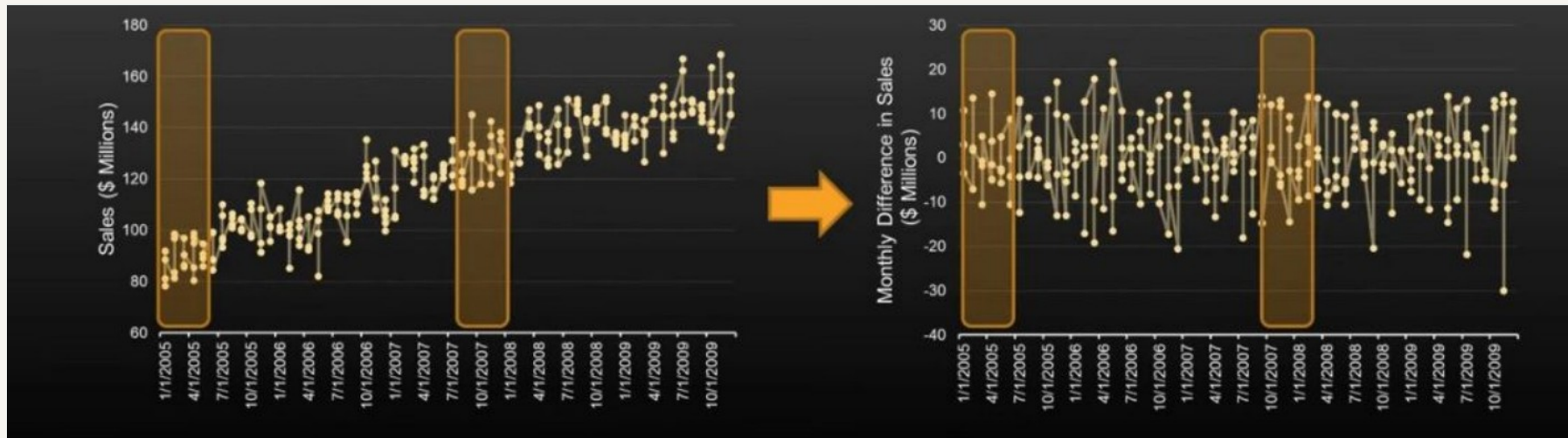
With stationary series the rejection rate stays near the nominal 5%

Remove a trend with the first difference $Y_t - Y_{t-1}$

$$\Delta Y_t = Y_t - Y_{t-1}$$

$$\Delta_S Y_t = Y_t - Y_{t-S}$$

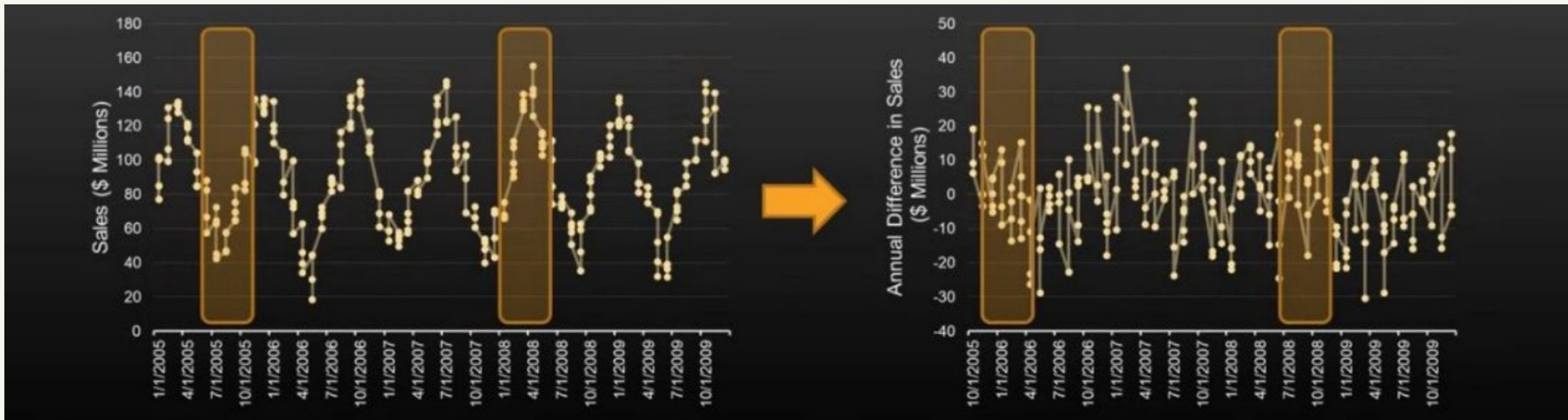
Lecture figure · monthly sales → monthly difference



After differencing, every window hovers around zero

Remove seasonality with the seasonal difference $Y_t - Y_{t-s}$

Lecture figure · monthly sales → year-over-year difference ($S = 12$)



MONTHLY

$S = 12$

QUARTERLY

$S = 4$

DAILY (weekly)

$S = 7$

NOTE

log before diff

Differencing is common, not the only fix — deterministic trends and seasonal dummies also work

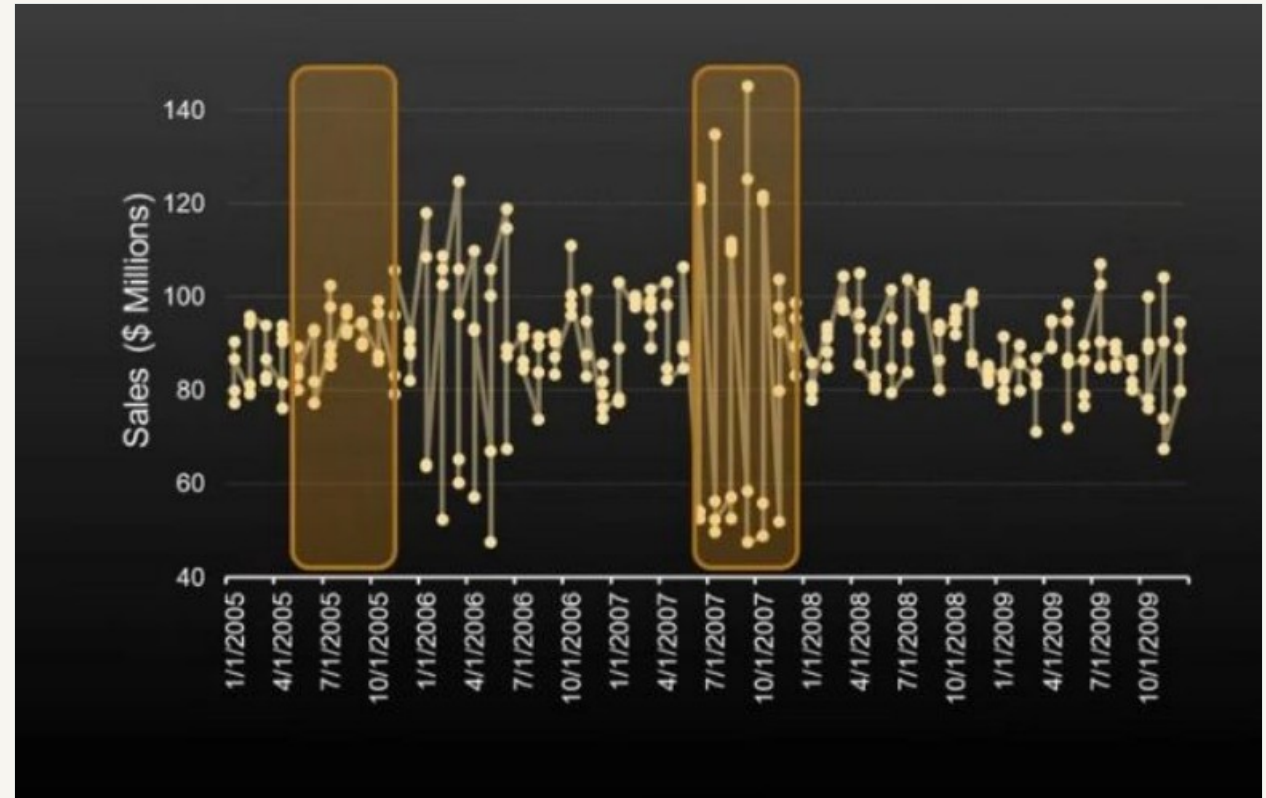
Differencing cannot fix a variance that shifts over time

LECTURE · 'Nope...'

Volatility shifts

- diff fixes the mean only
- ignoring it is 'LAZY'
- intervals go wrong

Fix: ARCH / GARCH



Mean models (ARIMA) and variance models (GARCH) do different jobs

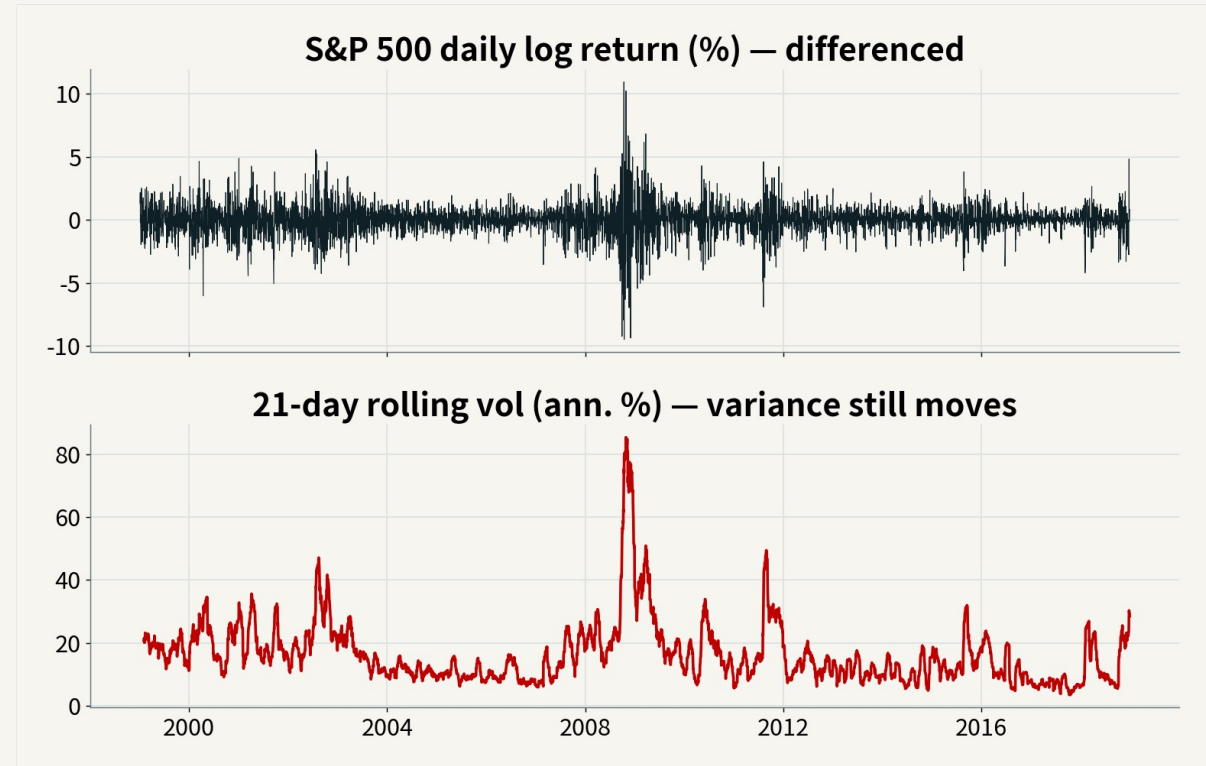
Stock returns are already differenced, yet volatility still clusters

GARCH(1,1) · Bollerslev (1986)

$$r_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

Volatility clustering

- big moves follow big moves
- β : persistence
- Engle, 2003 Nobel Prize



S&P 500 1999–2018: volatility bursts in 2008, 2011 and 2015

Model the mean with ARIMA and the variance with GARCH

Each kind of non-stationarity needs its own fix

VIOLATION	SYMPTOM	DIAGNOSIS	REMEDY
Trend	mean keeps moving	plot · ADF/KPSS	first difference d
Unit root	variance grows with t	ADF/KPSS	difference (d = 1)
Seasonality	repeats every S	seasonal ACF spikes	seasonal difference D
Vol. clustering	variance clusters	ACF of r^2_t	ARCH / GARCH

Only after stationarity can AR and MA be stacked on top

02

Autoregressive (AR) — explain with past values

The regressors are the series' own past

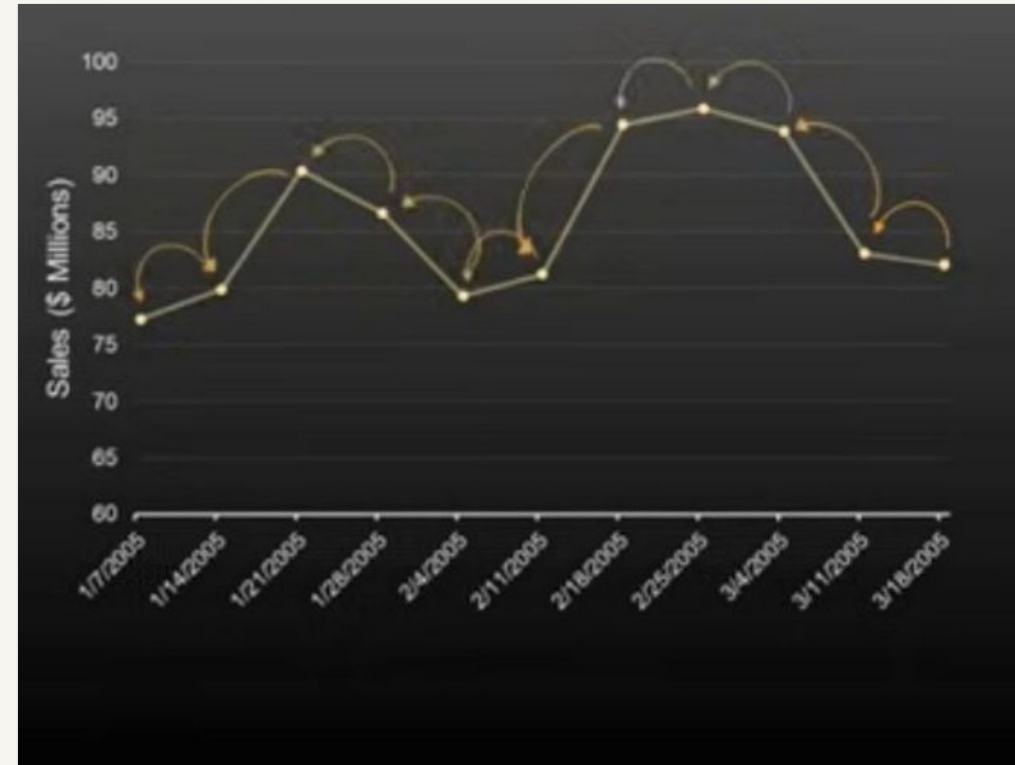
AR models forecast from the series' own past values — its lags

IDEA · lecture figure

Look back one step

- Y_{t-1} = lag-1 value
- regressor = own past
- hence 'auto'-regression

Each week uses last week



AR(1): depends on exactly one lag

Read AR(1) as four pieces: intercept, coefficient, lag, error

AR(1) model

$$Y_t = \omega + \phi Y_{t-1} + e_t$$

ω

Intercept

sets the mean

ϕ

Coefficient

persistence

Y_{t-1}

Lag value

last actual

e_t

Error (shock)

today's news

ϕ can be fit by OLS, but maximum likelihood is the norm

Substitute recursively and past shocks pile up — a 'long-memory' model

Plug in $Y_{t-1} = \omega + \phi Y_{t-2} + e_{t-1}$, then repeat

$$\begin{aligned} Y_t &= \omega + \phi Y_{t-1} + e_t \\ &= \omega + \phi(\omega + \phi Y_{t-2} + e_{t-1}) + e_t \\ &= \omega^* + \phi^2 Y_{t-2} + \phi e_{t-1} + e_t \\ &= \omega^* + \phi^3 Y_{t-3} + \phi^2 e_{t-2} + \phi e_{t-1} + e_t \end{aligned}$$

Lecture: 'This recursion goes back until the beginning of the series'
That is why AR is called a long-memory model

If $|\phi| < 1$, shocks from long ago almost vanish

AR(1) unrolled back to the first observation Y_1

$$Y_t = \omega \frac{1 - \phi^{t-1}}{1 - \phi} + \phi^{t-1} Y_1 + \sum_{j=0}^{t-2} \phi^j e_{t-j}$$

FIRST OBSERVATION

$$\phi^{t-1} \cdot Y_1$$

matters 'a little'

PAST SHOCKS

$$\phi^j \cdot e_{t-j}$$

older = smaller

CONDITION

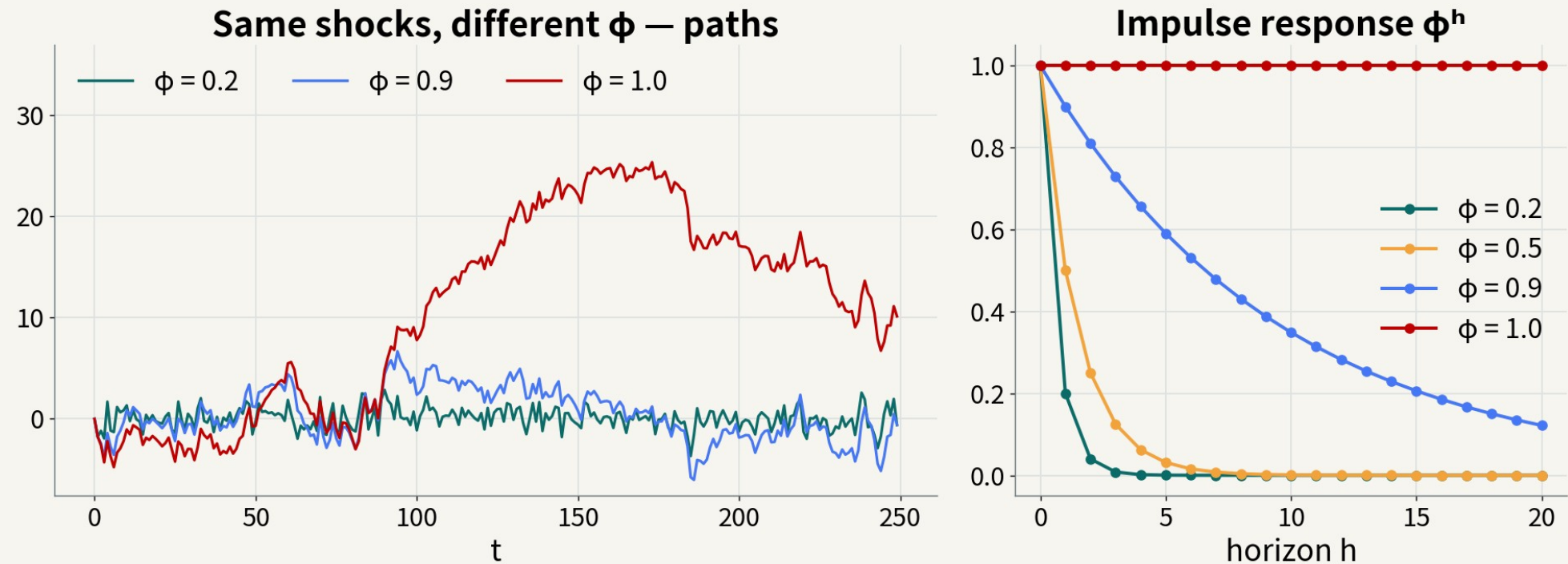
$$|\phi| < 1$$

mean-reverting

※ The lecture writes $\phi^t Y_1, \phi^{t-1} e_2$; starting from Y_1 the exact exponents are $\phi^{t-1} Y_1, \phi^{t-2} e_2$ (same conclusion)

With $\phi = 1$, shocks never fade — a random walk

The closer ϕ is to 1, the longer shocks last; $\phi = 1$ is a random walk



※ All three paths share the same shocks · right: what is left of a unit shock after h periods, ϕ^h

Stationary AR(1): ϕ sets the mean, variance and ACF

Moments of a stationary AR(1) ($|\phi| < 1$)

$$\mu = \frac{\omega}{1-\phi}, \quad \text{Var}(Y_t) = \frac{\sigma^2}{1-\phi^2}, \quad \rho(h) = \phi^h$$

MEAN LEVEL

$$\omega / (1 - \phi)$$

first term in the lecture

VARIANCE

$$\sigma^2 / (1 - \phi^2)$$

explodes as $\phi \rightarrow 1$

AUTOCORRELATION

$$\rho(h) = \phi^h$$

decays geometrically

A slowly decaying ACF is the fingerprint of AR

Add a second lag, or p lags, and you get AR(2) and AR(p)

AR(2) — two past values + error

$$Y_t = \omega + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + e_t$$

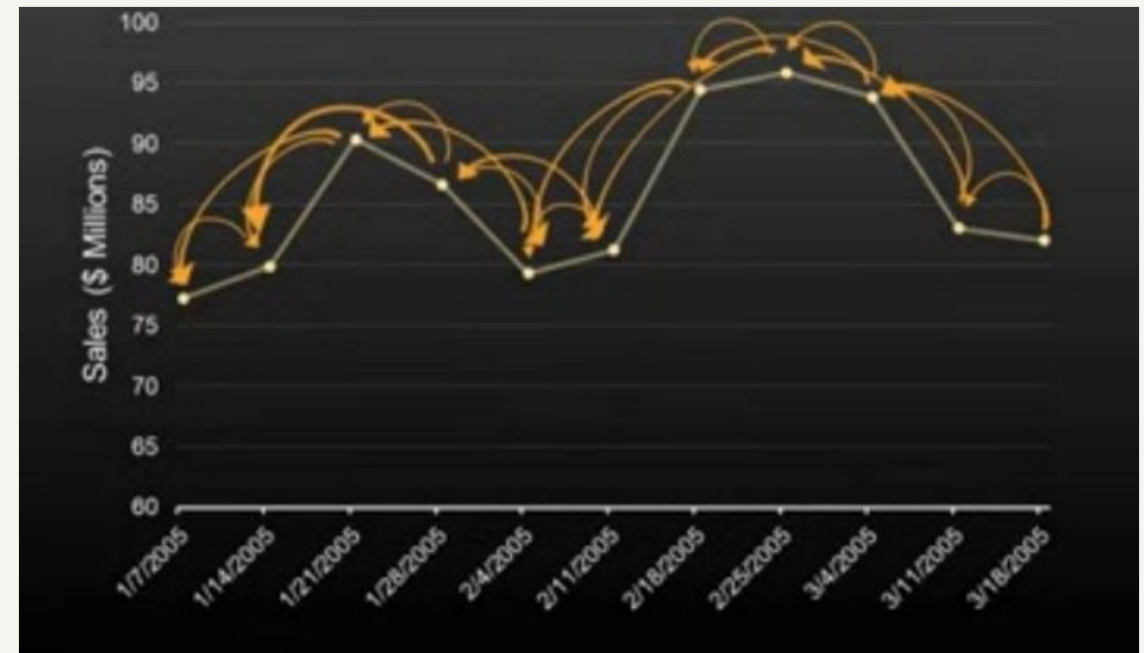
AR(p) — p past values + error

$$Y_t = \omega + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + e_t$$

Stationarity · all roots of the characteristic polynomial outside the unit circle

Each point now looks two steps back — twice the arrows

Lecture figure · 'So Many Lines!!!'



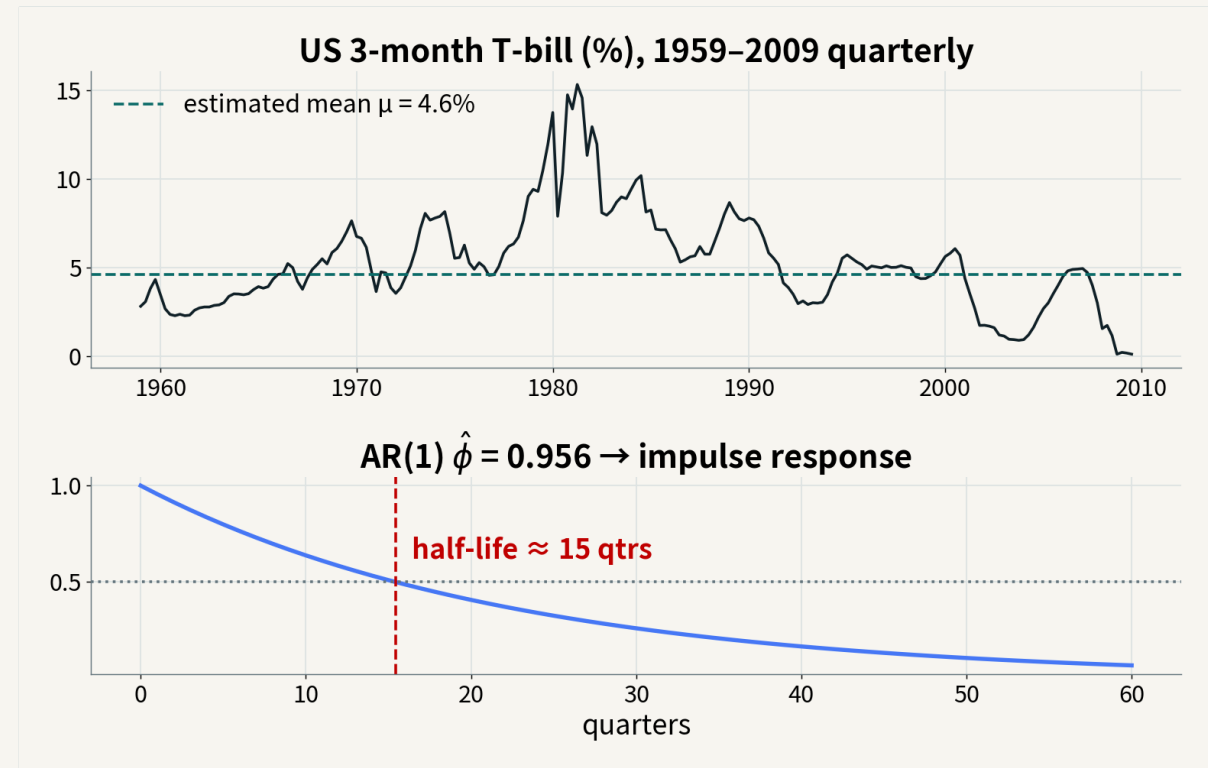
Interest rates revert to the mean — but very slowly

CASE · half-life of a T-bill shock

$$h_{1/2} = \frac{\ln 0.5}{\ln \hat{\phi}}$$

$$\hat{\phi} = 0.956$$

- half-life ≈ 15 qtrs (~ 4 yrs)
- discrete-time Vasicek
- ADF $p = 0.27$ (no reject)



When ϕ is this close to 1, AR(1) and a random walk are hard to tell apart

03

Moving Average (MA) — explain with past errors

'Come back average!' — shocks linger briefly, then the mean returns

MA swaps the regressor from past values to past errors

AR(1) · lagged target Y_{t-1}

$$Y_t = \omega + \phi Y_{t-1} + e_t$$

MA(1) · lagged error e_{t-1} ('Bait and Switch!')

$$Y_t = \omega + \theta e_{t-1} + e_t$$

WHAT IS THE ERROR

actual – forecast

$$e_t = Y_t - \hat{Y}_t$$

ROLE OF θ

correct by the miss

next forecast adds θe_t

CAUTION

MA $\neq$ rolling mean

not the chart indicator

Errors are unobserved, so MA is fit by maximum likelihood, not OLS

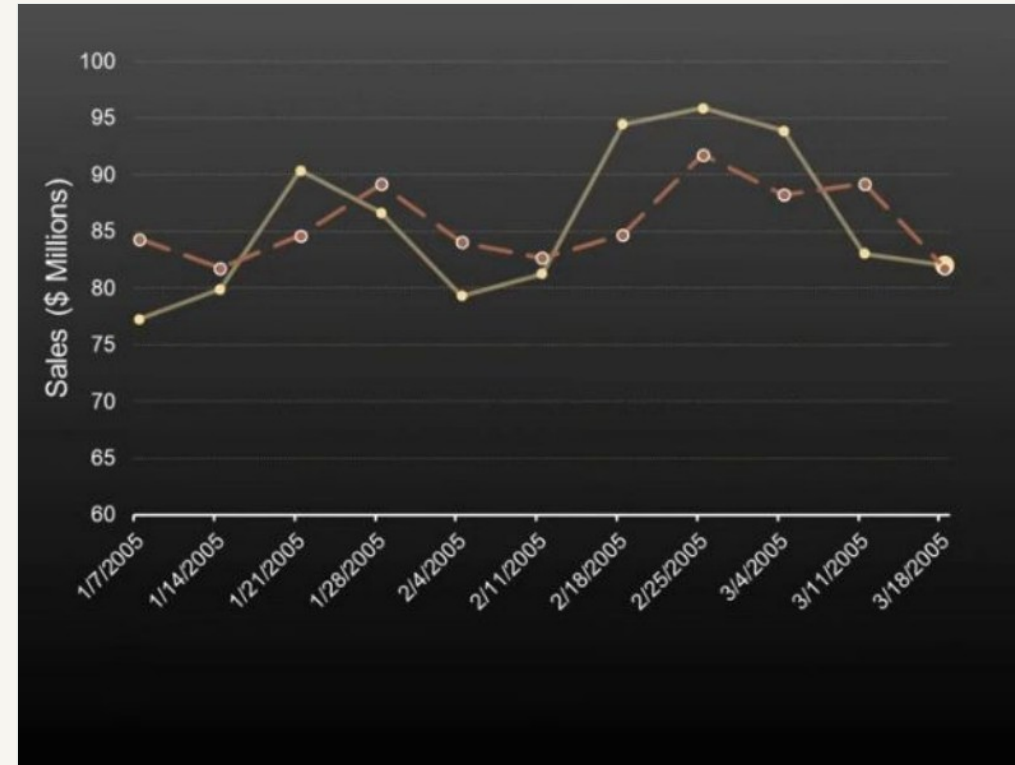
An MA(1) forecast only asks how far the last forecast missed

HOW TO READ · lecture figure

Solid actual, dashed fit

- the 'Wait... Huh?' chart
- dashed = MA(1) fit
- gap = the error

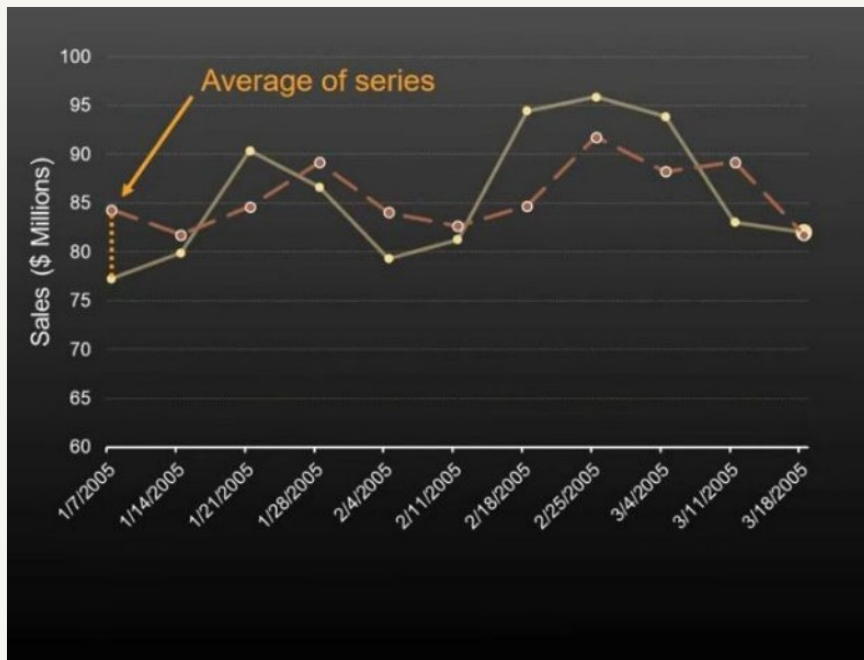
The fit trails by one step



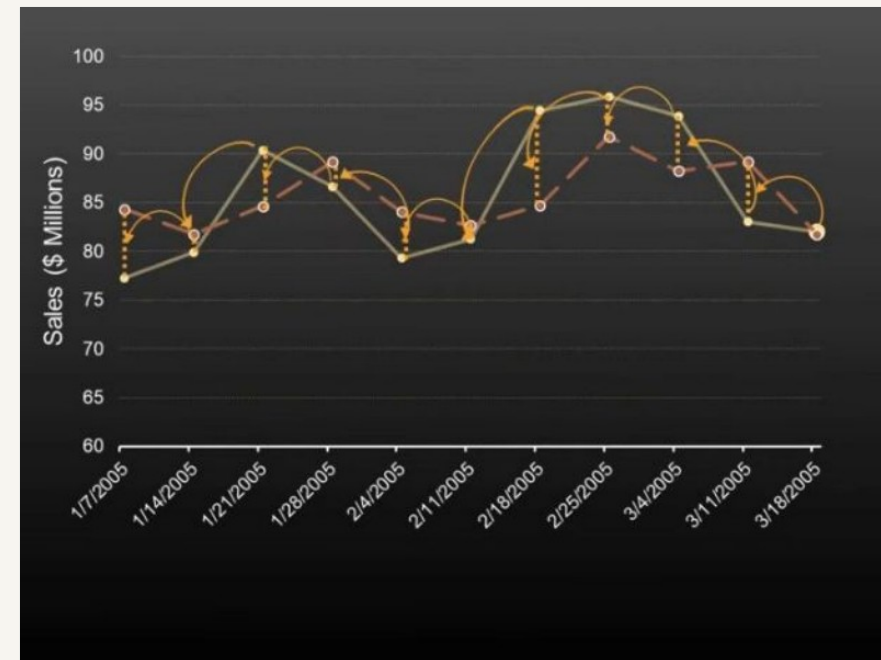
Big miss → big adjustment; small miss → stay near the mean

Every point corrects itself using the error one step back

① The first forecast starts at the series mean



② Then each step adjusts by the last error



'Everyone Looks Back One!' — AR looks at values, MA at errors

In MA(1) a shock drops out after one period — 'short memory'

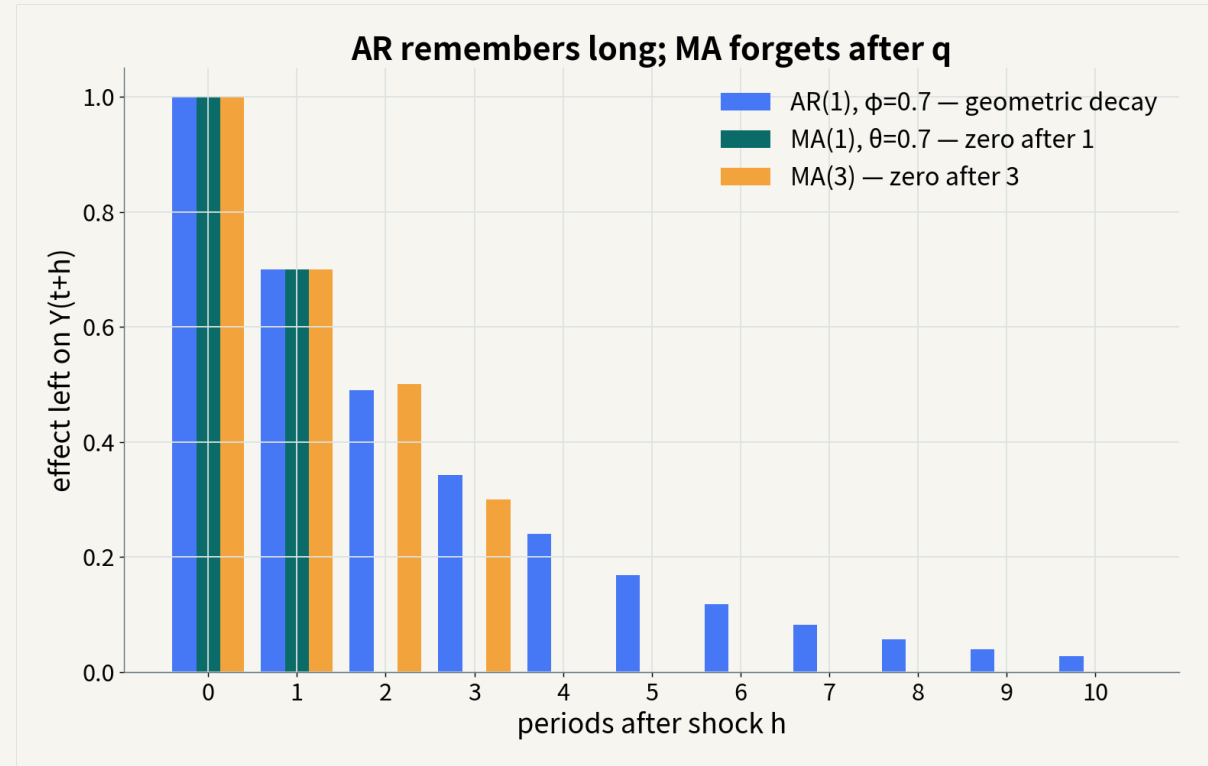
e_{t-1} is absent from Y_{t+1}

$$Y_t = \omega + \theta e_{t-1} + e_t$$

$$Y_{t+1} = \omega + \theta e_t + e_{t+1}$$

$$Y_{t+2} = \omega + \theta e_{t+1} + e_{t+2}$$

Lecture · 'So My Errors are Gone!!!'



AR forgets gradually (ϕ^h); MA forgets all at once after q

MA models are always stationary; their ACF cuts off at lag q

Mean and ACF of MA(1)

$$E[Y_t] = \omega, \quad \rho(1) = \frac{\theta}{1+\theta^2}, \quad \rho(h) = 0 \quad (h \geq 2)$$

Invertibility — MA(1) can be written as AR(∞)

$$|\theta| < 1 : e_t = \sum_{j=0}^{\infty} (-\theta)^j (Y_{t-j} - \omega) \Rightarrow \text{MA}(1) = \text{AR}(\infty)$$

STATIONARITY

holds for any θ

IDENTIFICATION

ACF zero after q

INVERTIBILITY

$|\theta| < 1 \rightarrow \text{unique}$

Lecture 'Two Different Ways': AR needs $|\phi| < 1$, MA needs nothing

AR(p) and MA(q) differ in what they remember

MA(q)

$$Y_t = \omega + \theta_1 e_{t-1} + \theta_2 e_{t-2} + \cdots + \theta_q e_{t-q} + e_t$$

AR(p)

$$Y_t = \omega + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + e_t$$

ASPECT	AR(p)	MA(q)
Regressors	past values $Y_{t-1} \dots Y_{t-p}$	past errors $e_{t-1} \dots e_{t-q}$
Memory	long · geometric decay	short · zero after q
Stationary if	roots outside unit circle	always
ACF / PACF	tails off / cuts at p	cuts at q / tails off

Their ACF and PACF shapes are mirror images

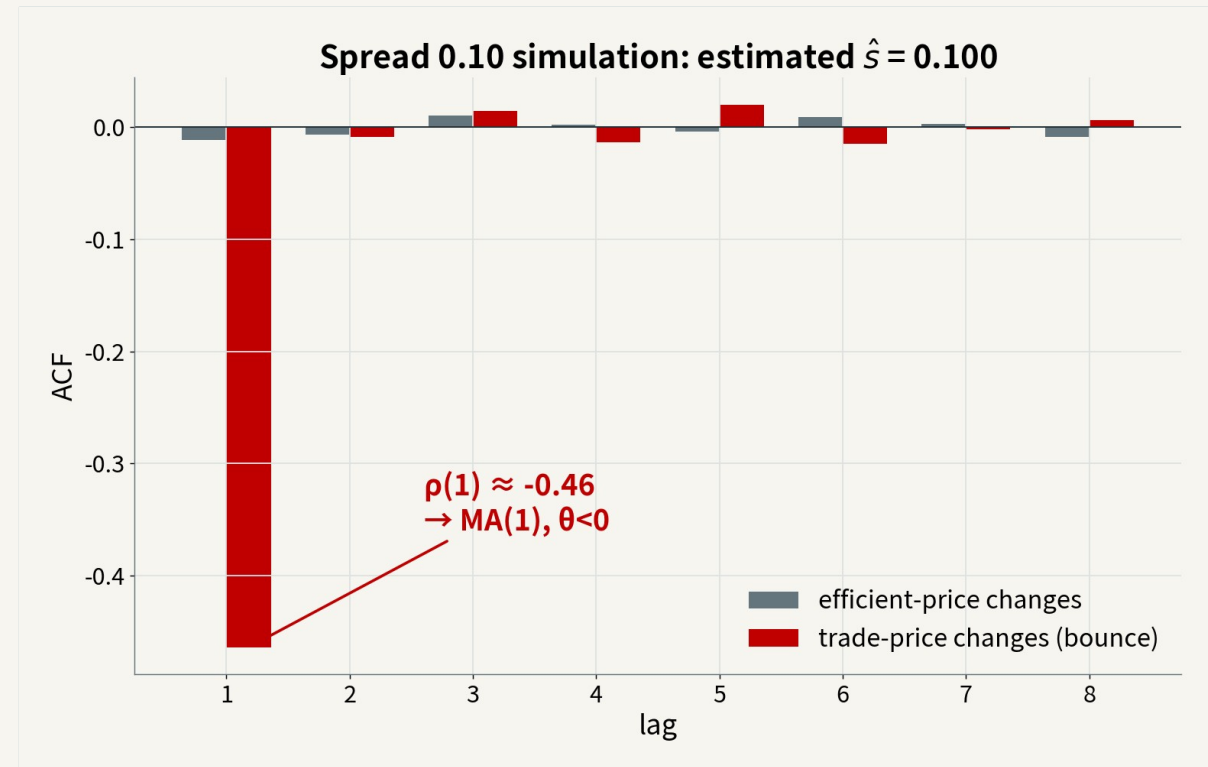
Bid-ask bounce plants a negative MA(1) in trade-price returns

CASE · Roll (1984) spread estimator

$$\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -\frac{s^2}{4} \Rightarrow \hat{s} = 2\sqrt{-\widehat{\text{Cov}}}$$

Bid-ask bounce

- simulated $\rho(1) \approx -0.46$
- spread 0.10 $\rightarrow$ est. 0.100
- typical of intraday data



MA structure arises naturally when prices carry measurement noise

04

ARMA · ARIMA — combine, then difference

'Can't Believe All These Videos Tie Together...'

AR(1) equals an infinite-order MA — a pile of short memories

$$\text{AR}(1) = \text{MA}(\infty) \quad (|\phi| < 1)$$

$$Y_t = \mu + e_t + \phi e_{t-1} + \phi^2 e_{t-2} + \cdots = \mu + \sum_{j=0}^{\infty} \phi^j e_{t-j}$$

LECTURE

'Large MA term!!'

recursion tail = big MA

READING

Summed shocks

weighted by ϕ^j

GENERAL · WOLD

stationary = MA(∞)

any stationary series

Mixing them lets a few terms stand in for very long AR or MA

ARMA(p, q) uses past values and past errors together

ARMA(p, q)

$$Y_t = \omega + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j e_{t-j} + e_t$$

p

AR order

how many past values

q

MA order

how many past errors

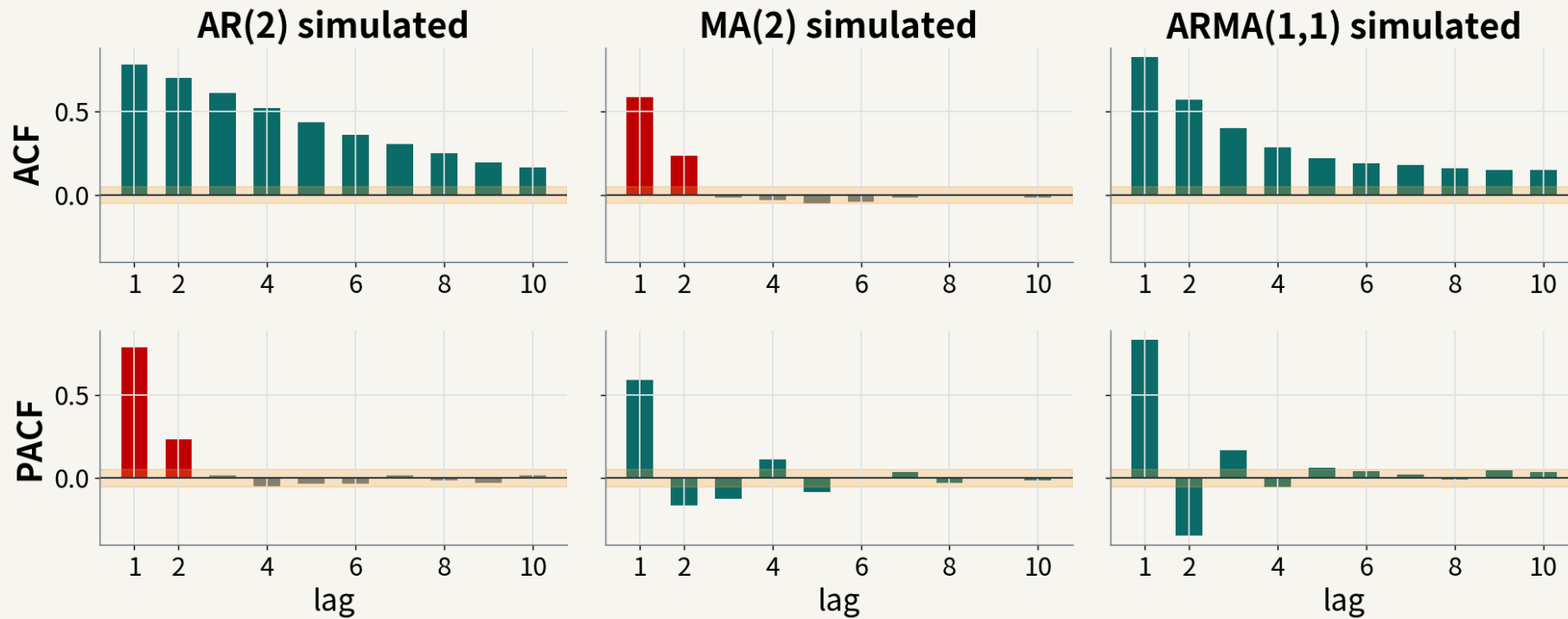
PARSIMONY

one ARMA(1,1)

replaces long AR or MA

The real question: how many p's and q's?

Use where ACF and PACF cut off to gauge p and q



※ 1,500 simulated points · red bars = cut-off, band = $\pm 1.96/\sqrt{n}$

The side that cuts off tells you the order

MODEL	ACF	PACF (partial)	READ THE ORDER
AR(p)	tails off	cuts off after p	p from PACF
MA(q)	cuts off after q	tails off	q from ACF
ARMA(p,q)	tails off	tails off	use IC / auto selection

※ PACF(h): correlation of Y_t and Y_{t-h} after removing lags $1 \dots h-1$ — equals the last AR(p) coefficient

If both tail off, plots are not enough → automatic selection

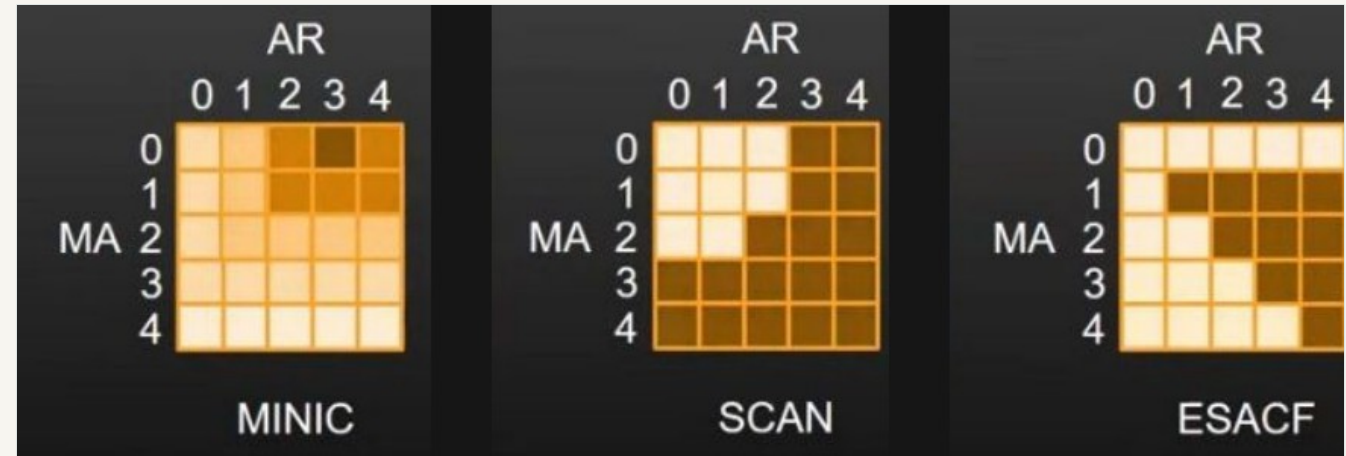
Automatic selection picks candidates from a (p, q) grid

LECTURE · SAS PROC ARIMA

Three auto methods

- MINIC: min-BIC cell
- SCAN: corner of zeros
- ESACF: zero-triangle vertex

Bright cells = candidates



Start estimating where the three methods agree

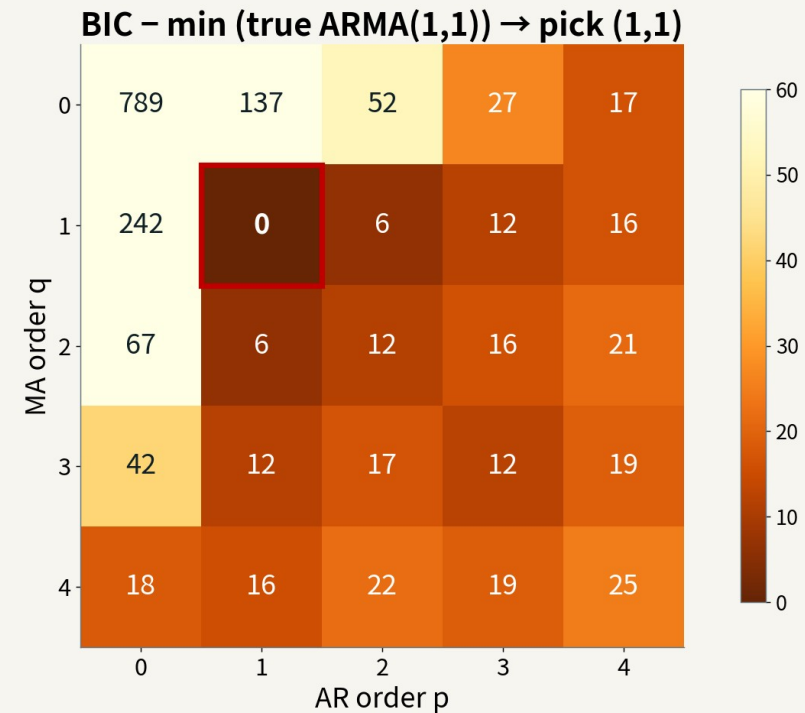
Information-criterion grids balance fit against the number of parameters

k = # parameters, n = sample size

$$AIC = -2 \ln \hat{L} + 2k, \quad BIC = -2 \ln \hat{L} + k \ln n$$

True (1,1) recovered

- AIC: forecast-oriented
- BIC: favours parsimony
- auto_arima automates it



The grid narrows the field; residual checks make the final call

The I in ARIMA means 'integrated' — fit ARMA to the differenced series

ARIMA(p, d, q) with the backshift operator B

$$\phi(B) (1 - B)^d Y_t = \omega + \theta(B) e_t, \quad B Y_t = Y_{t-1}$$

INTEGRATED

inverse of diff = sum

I(1) = summed once

MODELLING

difference d times

then fit ARMA

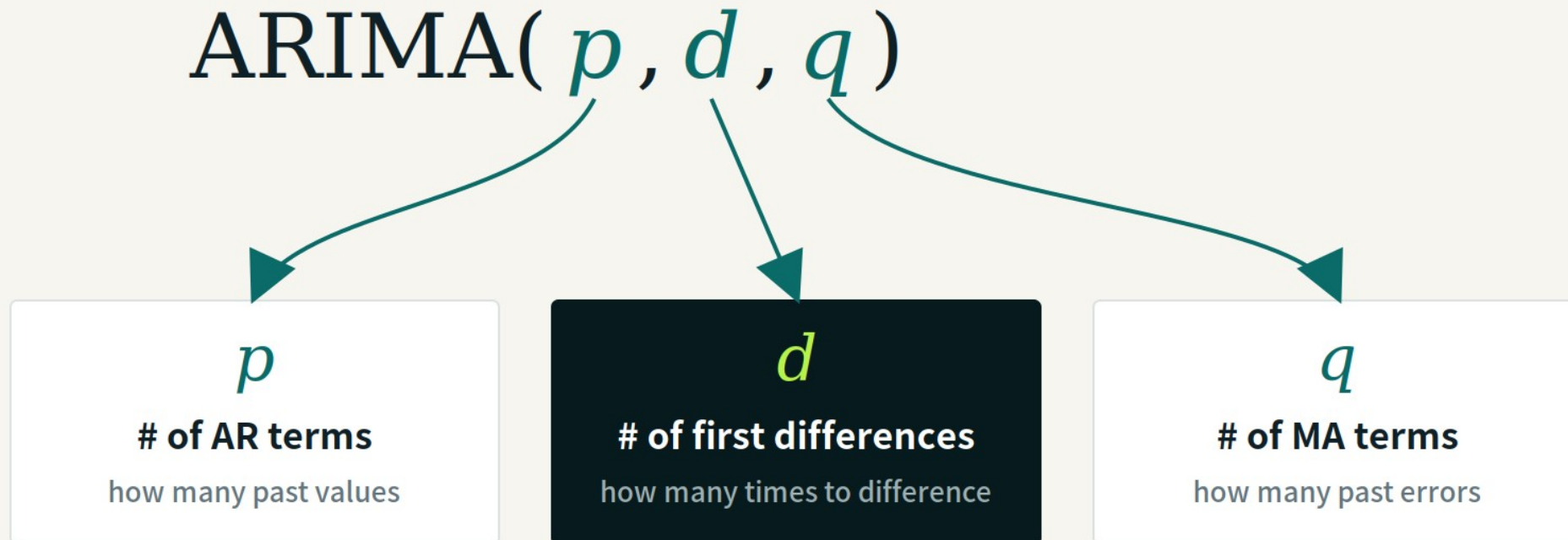
FORECAST

integrate back

to the original scale

Lecture: 'make your data stationary (typically through differencing)'

ARIMA(p, d, q): three numbers sum up the model



※ $\text{ARIMA}(p, 0, q) = \text{ARMA}(p, q)$ · $\text{ARIMA}(0, 1, 0) = \text{random walk}$ · $\text{ARIMA}(0, 1, 1) = \text{simple exponential smoothing}$

ARIMA(1,1,1): difference once, then one AR and one MA term

Lecture 'Time to Wrap Up!!!'

$$W_t = Y_t - Y_{t-1}$$

$$W_t = \omega + \phi_1 W_{t-1} + \theta_1 e_{t-1} + e_t$$

d = 1

$$W_t = Y_t - Y_{t-1}$$

difference first

p = 1

$$\phi_1 W_{t-1}$$

lag of the differenced

q = 1

$$\theta_1 e_{t-1}$$

last error

The AR and MA terms act on W_t , not on the original Y_t

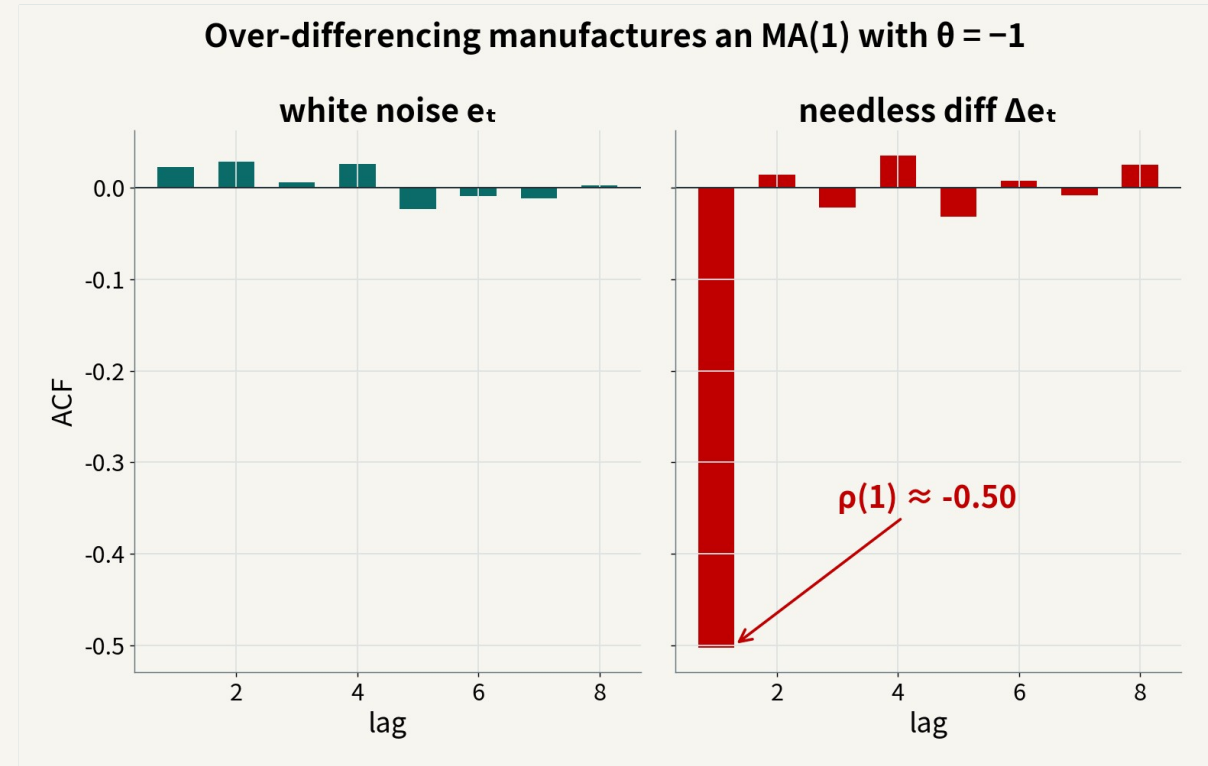
Difference only as much as needed — over-differencing fakes an MA

Differencing white noise again

$$\Delta e_t = e_t - e_{t-1} \Rightarrow \rho(1) = \frac{-\sigma^2}{2\sigma^2} = -0.5$$

d is usually 0 or 1

- set d with ADF/KPSS
- $\rho(1) \approx -0.5$: red flag
- $\hat{\theta} \approx -1$: too much



Over-differencing inflates variance and hurts forecasts

Box–Jenkins: stationarize → identify → estimate → diagnose → forecast

The Box–Jenkins (1970) iterative cycle



※ Ljung–Box tests whether residual autocorrelations are jointly zero — if rejected, re-identify

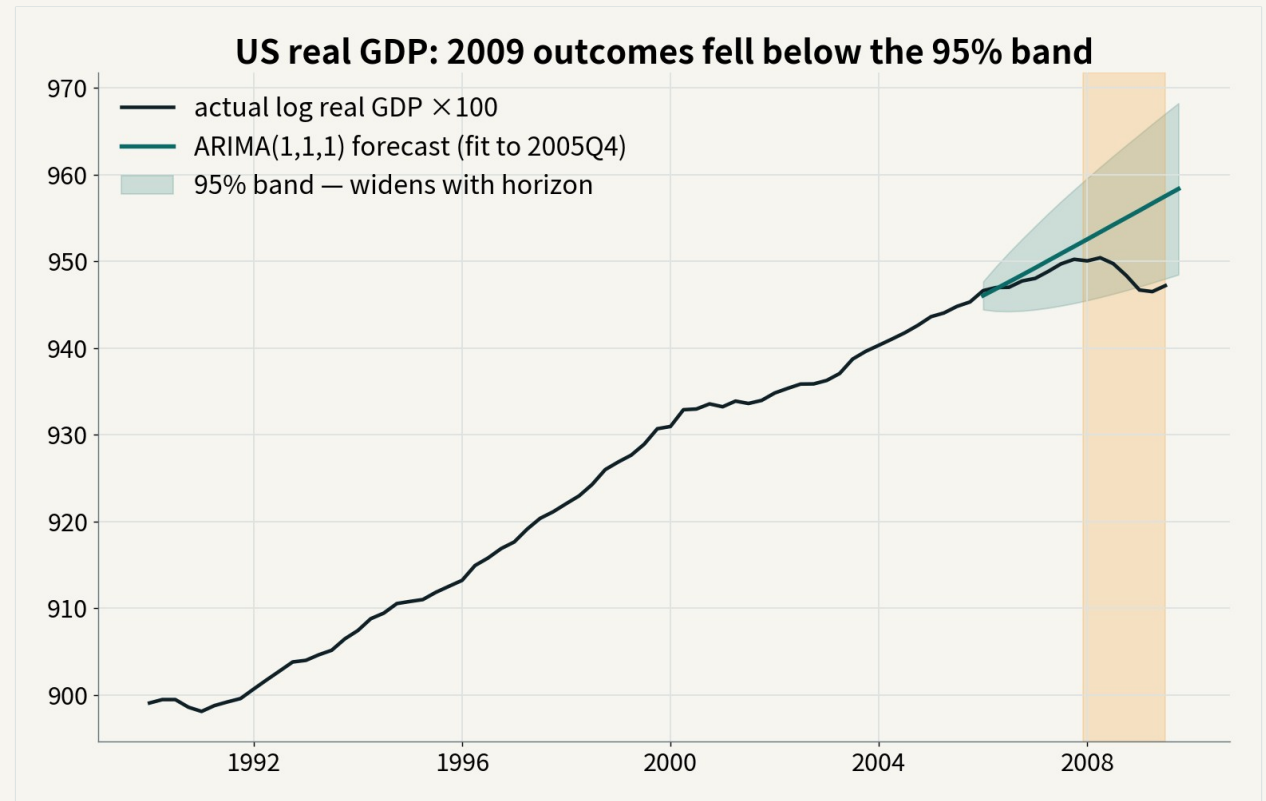
ARIMA forecast bands only capture 'normal-times' uncertainty

CASE · US REAL GDP

ARIMA(1,1,1)

- $\hat{\phi} = 0.64, \hat{\theta} = -0.41$
- band widens with horizon
- 2009 fell outside

Band exit = regime signal



Fit to 2005Q4 · 16-quarter forecast · shaded = 2007Q4–2009Q2 recession

Beating a simple benchmark is the first hurdle

EPISODE · 1983

Meese–Rogoff

- structural FX models
- lost to a random walk out of sample
- a long-standing puzzle

EPISODE · 1982–

M-Competitions

- Makridakis forecasting contests
- complex $\neq$ always better
- combinations do well

EPISODE · 2018

M4

- winner: ES + neural hybrid
- pure ML alone lagged
- statistical models still core

Always keep a random walk (ARIMA(0,1,0)) and a seasonal naive as benchmarks
Compare models only on out-of-sample performance

05

Seasonal ARIMA — look one season back

'Uh oh... Different Data' — data with a seasonal nature

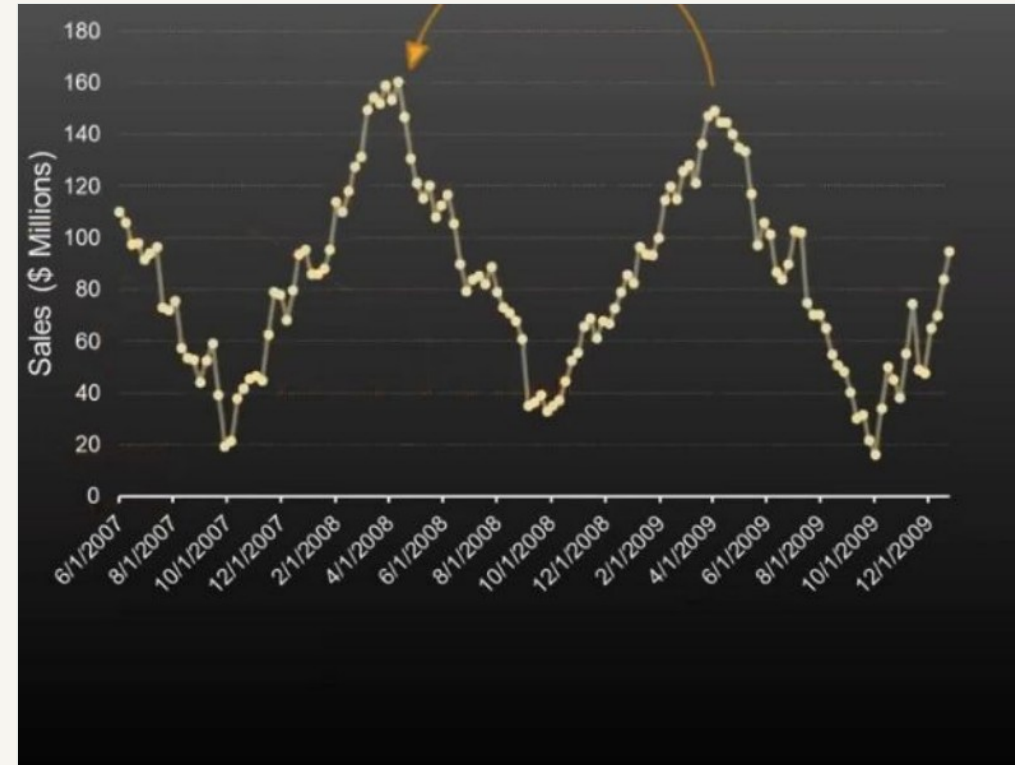
With seasonal data, one season ago also informs today

IDEA · lecture figure

Last month + last year

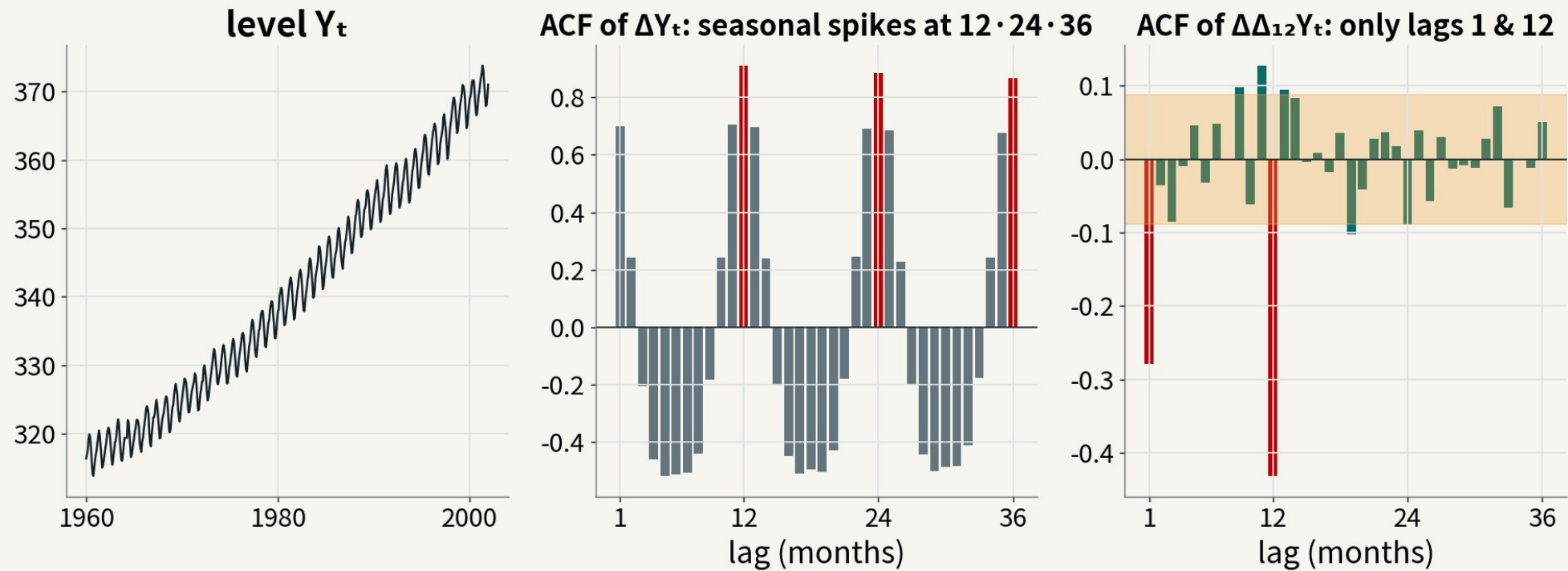
- monthly 12, quarterly 4
- spring '09 ← spring '08
- retail · power · CO₂

Give the season its own terms



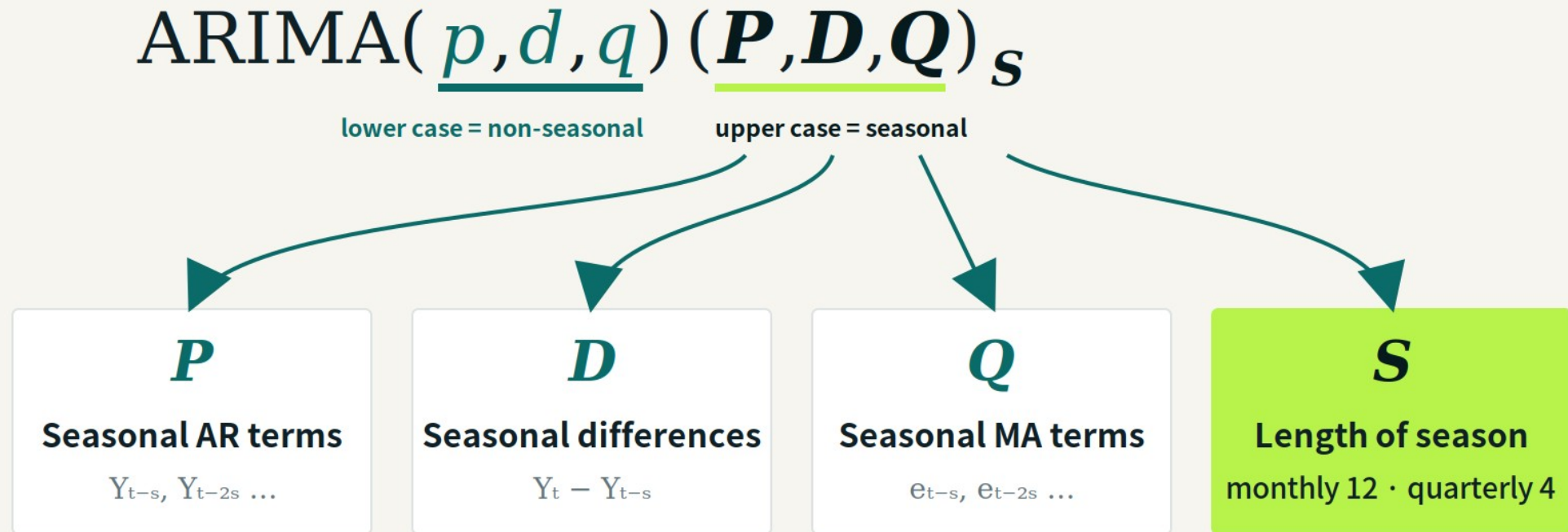
Recent lags alone cannot carry a 12-month cycle

Seasonality shows up as ACF spikes at lags 12, 24, 36



※ Mauna Loa CO₂ monthly (1960–2001) · after both differences only lags 1 and 12 remain → airline-model signature

Upper-case (P, D, Q)_s is the seasonal part



※ Lecture: 'Little letters are non-seasonal letters! Big letters are seasonal letters!'

SARIMA multiplies non-seasonal and seasonal polynomials

General SARIMA(p,d,q)(P,D,Q)_s

$$\phi_p(B) \Phi_P(B^S) (1 - B)^d (1 - B^S)^D Y_t = \omega + \theta_q(B) \Theta_Q(B^S) e_t$$

BACKSHIFT

$$B^s Y_t = Y_{t-s}$$

one season back

SEASONAL POLY.

$$\Phi(B^s), \Theta(B^s)$$

AR and MA at step S

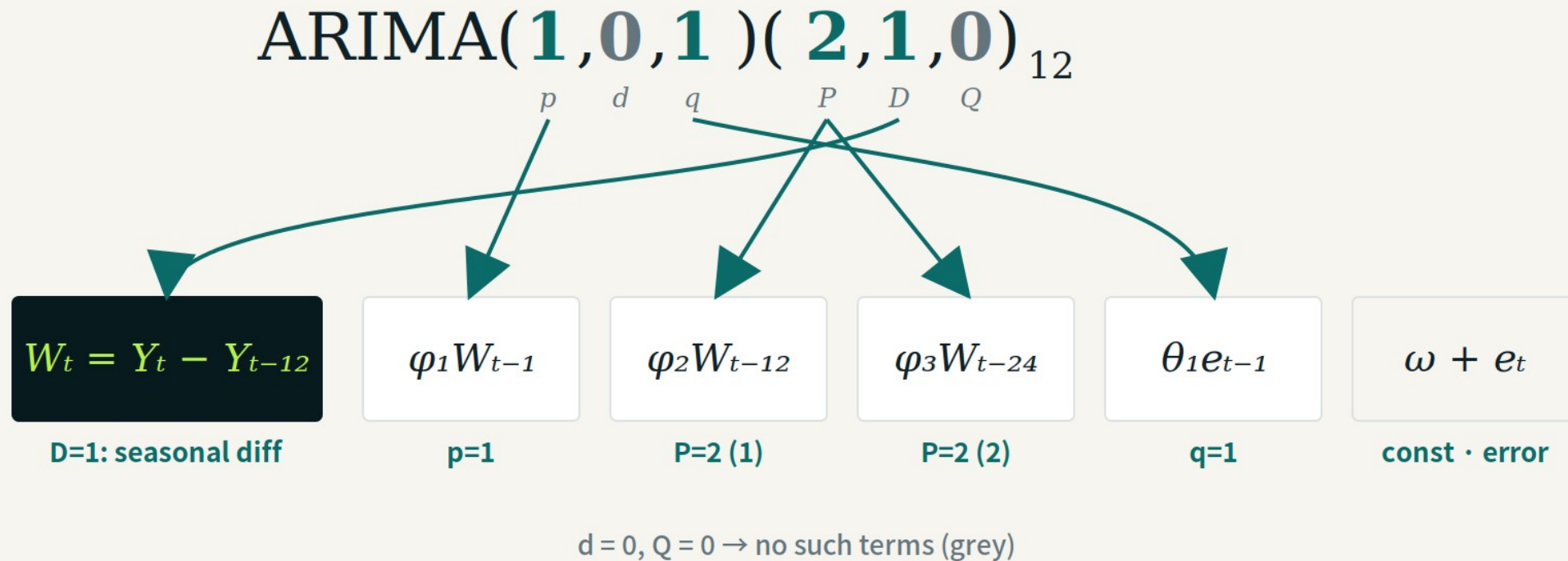
MULTIPLICATIVE

product of the two

creates cross terms

Apply $(1-B)^d(1-B^s)^D$ first, then catch what is left with ARMA

ARIMA(1,0,1)(2,1,0)₁₂: seasonal difference first — 'Differences First!'



※ Mapping follows the lecture: the two seasonal AR terms (ϕ_2, ϕ_3) are Φ_1, Φ_2 in standard notation

Standard software is multiplicative, so cross terms W_{t-13} and W_{t-25} appear

Lecture's additive form (a teaching simplification)

$$W_t = Y_t - Y_{t-12}$$

$$W_t = \omega + \phi_1 W_{t-1} + \phi_2 W_{t-12} + \phi_3 W_{t-24} + \theta_1 e_{t-1} + e_t$$

Multiplicative form (statsmodels · R · SAS)

$$(1 - \phi_1 B)(1 - \Phi_1 B^{12} - \Phi_2 B^{24}) W_t = \omega + (1 + \theta_1 B) e_t$$

$$W_t = \omega + \phi_1 W_{t-1} + \Phi_1 W_{t-12} - \phi_1 \Phi_1 W_{t-13} + \Phi_2 W_{t-24} - \phi_1 \Phi_2 W_{t-25} + \theta_1 e_{t-1} + e_t$$

Read coefficients in the multiplicative form, cross terms included

Read seasonal orders from ACF/PACF at multiples of S

PATTERN (S = 12)	MEANING	CHOOSE
ACF decays slowly at 12 · 24 · 36	seasonal non-stationarity	D = 1
ACF spike at 12, then cut	seasonal MA	Q = 1
PACF spikes at 12 · 24, then cut	seasonal AR	P = 2
after $\Delta\Delta_{12}$ only lags 1 · 12 (neg.)	Airline	$(0,1,1)(0,1,1)_{12}$

Read non-seasonal orders at lags 1–11 and seasonal ones at multiples of 12

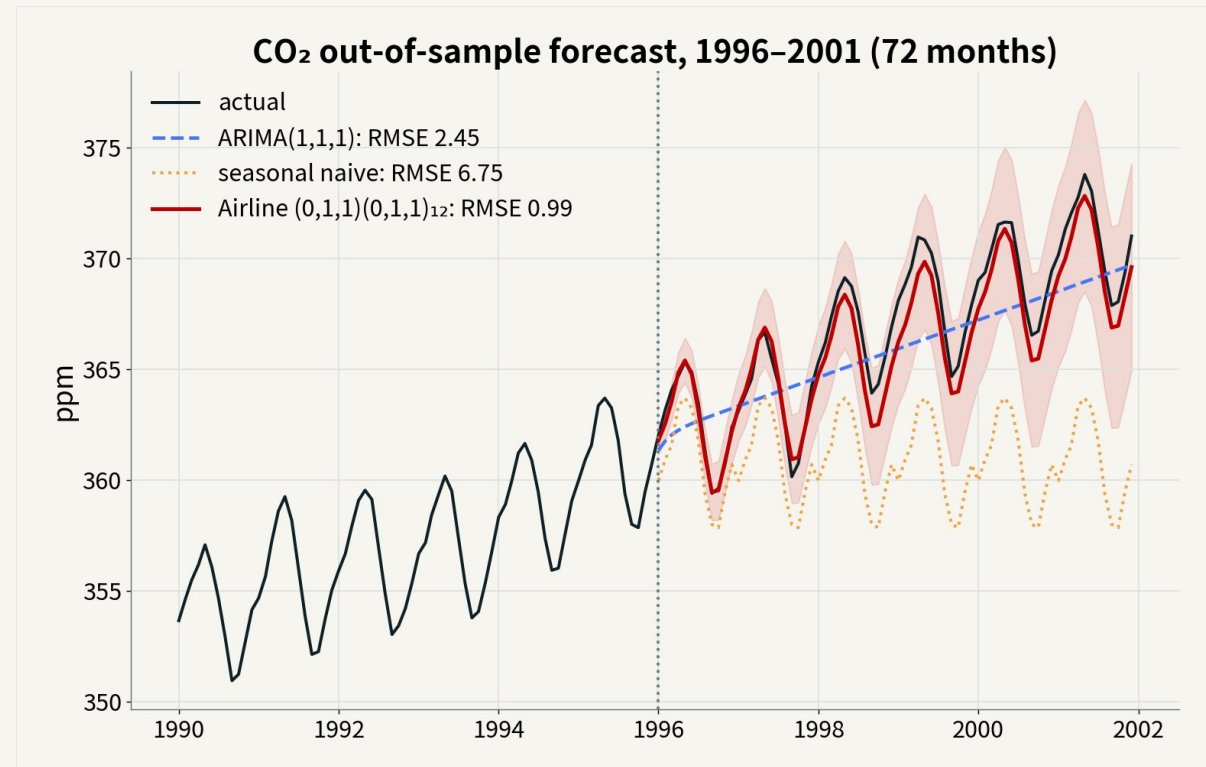
One airline model keeps 6-year CO₂ forecast error near 1 ppm

CASE · airline model on CO₂

$$(1 - B)(1 - B^{12})Y_t = (1 + \theta_1 B)(1 + \Theta_1 B^{12})e_t$$

RMSE 0.99 ppm

- $\hat{\theta}_1 = -0.37, \hat{\Theta}_1 = -0.85$
- ARIMA(1,1,1): 2.45
- seasonal naive: 6.75



The name comes from the airline passenger data in Box–Jenkins (1970)

Two parameters capture trend and season together — that is its power

Official seasonal adjustment runs on a SARIMA engine

OFFICIAL STATISTICS

X-13ARIMA-SEATS

- US Census Bureau program
- used for Korean statistics too
- regARIMA extends, then filters

AUTOMATION

auto_arima

- pmdarima · R forecast
- unit-root tests + IC search
- always diagnose the output

PITFALLS

Calendar & breaks

- moving holidays (e.g. Lunar New Year)
- trading-day counts
- handle as regressors

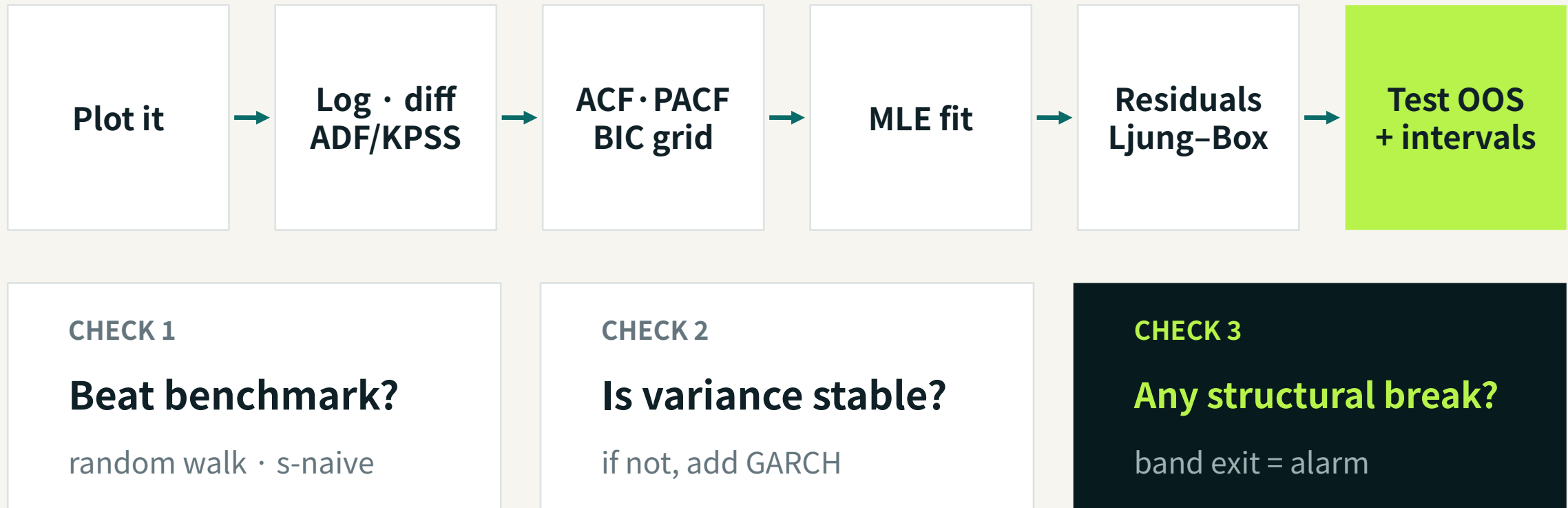
Behind every seasonally adjusted q/q figure sits a SARIMA forecast

The five models differ in what they use and what they fix first

MODEL	USES	MEMORY	SIGNAL	PREREQUISITE
AR(p)	past values	long	PACF cut at p	$ \phi < 1$
MA(q)	past errors	short	ACF cut at q	invertible
ARMA(p,q)	both	mixed	both tail off	stationary
ARIMA(p,d,q)	diff + ARMA	mixed	unit-root test	d differences
SARIMA	+ seasonal lags	seasonal	spikes at kS	D seasonal diffs

Every model aims at the same end: stationary input, white-noise residuals

In practice, check in this order



Data checks and validation design prevent more errors than model choice

Next: extend to variance, multiple series and non-linearity

VARIANCE

ARCH · GARCH

- model volatility clustering
- feeds VaR and option pricing
- ARIMA-GARCH combo

MULTIVARIATE

VAR · cointegration

- joint dynamics
- the right fix for spurious regression
- basis of pairs trading

BEYOND

State space · ML

- Kalman filter estimates SARIMA
- regime switching · HMM
- ARIMA as the ML baseline

ARIMA is not the end — it is the baseline for every time-series model

Stationarity is the premise,

AR and MA are the ingredients,

ARIMA and SARIMA are the recipe.

**Identify → estimate → diagnose
until the residuals are white noise**

Source: Aric LaBarr lecture slides (stationarity, AR, MA, ARIMA, seasonal ARIMA) · added charts reproduced with statsmodels datasets (macrodata, CO₂), arch S&P 500 and simulations